

☒ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. Project No.: WW-4705R
Project Name: Hilo WWTP Phase 1 Submittal Number: 02667-001.2
Submittal Title: Hot Tapping and Line Stopping ☐ For Information Only
TO: County
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec: 02667	Paragraph: 1.02
Authorized By: Koppl Pipeline Services, Inc.	Date Submitted: 09/09/2025

Submittal Certification	
Check Either (A) or (B):	
☐ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .
☒ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: <u>Nitesh N</u>	
Printed Name and Title: Nitesh Nakirikanti, Project Engineer	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/10/2025
SPECIFICATION SECTION # 02667
SPECIFICATION 1.02
PARAGRAPH 1.02
DRAWING N/A
SUBCONTRACTOR N/A
SUPPLIER Koppl Pipeline Services, Inc.
MANUFACTURER N/A

CERTIFIED BY CQCM or Designee: [Signature]

Review Comments:

1. Please verify that the contractor has inspected the pipe where the taps will be made and the line stop is to be installed and that the materials and procedures submitted are based on the findings.

Incorporated - contractor has inspected the pipe

2. The submitted hot tap procedure is acceptable. Note that specification 02d667, 3.03A requires a four hour hydrostatic test of the tapping sleeve and valve prior to tapping the pipe.

Incorporated - 4 hour testing can be done

1. After the hot taps are complete, please deliver the coupons to the engineer rather than disposing of them.

Incorporated - coupons will be delivered to the engineer

4. Specification 02667, 2.01A requires a Mueller H-615 tapping sleeve. If the contractor wishes to use another type or brand, please explain the reasons and benefits.

Response - The vendor has indicated that the Mueller H-615 comes with a substantial lead time. They recommend an alternative JCM that they have successfully used on multiple past projects and feel more comfortable moving forward with, also meets safety and pressure requirements.

5. The Mueller 12-inch flange by flange A-2360 gate valve is acceptable except that per specification 02667, 2.01 C2 the shaft/stem is to be stainless steel.

Incorporated - shaft/stem will be SST

6. What is the purpose and location of the small ball valve?

Incorporated - The small hot tap is for a control point to blowdown when the line stop is being set and for equalization to remove the line stop.

7. The calculations are Filled for Record. They should be stamped and signed by the engineer who made them.

Incorporated

REQUEST FOR INFORMATION (RFI) RESPONSE

Owner: County of Hawai'i **Project No.:** 203795
Contractor: Nan, Inc. **RFI No.:** 0025
Project Name: Hilo WWTP Rehabilitation and Replacement
Project Phase 1 **Spec/Dwg
Reference:** 02667
RFI Title: Clarification on Hot Tapping and Line Stopping Spec Requirements

Information Requested

We are seeking clarification on a few points raised by our hot tapping and line stopping vendor, Koppl, regarding Section 02667 – Hot Tapping and Line Stopping. Please find the specific items requiring clarification below:

1. Submittals

1.1 Section 1.02 – G: Backup Equipment

Koppl's Concern: They do not typically provide backup equipment; their standard practice is to stop and seal the line if needed.

Request: What types of backup equipment are expected, or can they operate as per their standard practice?

1.2 Section 1.02 – H: Design Calculations for Components

Koppl's Concern: They need clarification on "draw structure" and "pressure plate" components.

Request: Could you provide definitions and clarification on these terms and the required calculations?

1.3 Section 1.02 – I: Single Supplier Requirement

Koppl's Concern: They typically source components from multiple suppliers.

Request: Can this requirement be flexible, or can they list suppliers per component?

2. Maximum Operating Pressure

Koppl's concern: Define the maximum operating pressure as referred to in section 3.03 - A

Please provide clarification to help Koppl move forward.

Authored

By: Nan, Inc.

Date

Submitted: 05/14/2025

Response to Request:

1.1) Section 1.02-G, Back-up Equipment

The back-up equipment required should be based on the equipment the contractor proposes to use to complete the work and which, if a failure occurs, could delay successful and timely completion of the hot taps and the line stop and/or could result in leakage of wastewater and hazardous materials. Depending on the type of equipment the contractor will use, this could include power supply, hydraulic pump, hoses and fluids, and spare parts including bolts, gaskets and cutters.

1.2) Section 1.02-H, Design Calculations

Disregard the terms "draw structure" and "pressure plate."

1.3) Section 1.02-I, Single Supplier Requirement

The intention of this section is to ensure that, under the general contractor, there is a single source of responsibility for providing the complete hot tapping and line stopping operation including supplying components and equipment and completing the installation and testing. In this case, that source is likely the specialty subcontractor retained by the general contractor.2. Contractor and their subcontractor shall provide a list of all pipes that are being hot-tapped; pressure for these pipes will be provided by Engineer.

2) Section 3.03-A, Maximum Operating Pressure

Use 60 psi for the maximum operating pressure for the required test of the tapping sleeves and valves. The test pressure for these components will therefore be 120 psi. Please note that this is not the test pressure to be used for testing the entire system after the line stop has been installed.

In the event, Contractor believes the RFI response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.

Firm
Name: Carollo Engineers

Date
Returned: 6/5/2025

Signature: Jim Ewing, Walid Karam

END OF DOCUMENT

SECTION 02667

HOT TAPPING AND LINE STOPPING

PART 1 GENERAL

1.01 SUMMARY

- A. Section Includes: This section includes materials and installation of hot-tap connections to existing ductile-iron pipelines with line stop. All equipment and specialized appurtenances shall be provided by one supplier.

1.02 SUBMITTALS

- A. All submittals shall be made in accordance with Section 01330 - Submittal Procedures.
- B. Experience summary and references for specialty contractor to perform the work.
- C. Step-by-step description of procedures to be used for hot tapping and line stopping. A contingency plan shall be developed and submitted for each step considered to place the plant operation at risk of being out of service.
- D. Shop drawings for all materials used in the hot tapping and line stopping of the pipeline.
- E. Installation procedures including tapping sleeves and pressure plate, tapping nozzle and appurtenances, and line stop. Submittal shall include procedures for hot tap installation, removing pipe wall coupon, installing fabricated sleeve type couplings, tie-in of tapping nozzle to new pipeline, and installation and later removal of line stop.
- F. Installation and removal procedures for the line stop.
- G. List of equipment to be used in the installation with demonstration of backup equipment available to be on site prior to commencement of work.
- H. Calculations for design of tapping sleeves, draw structure, pressure plate, and line stop.
- I. Letter verifying that all equipment and specialized appurtenances are provided by one supplier.

1.03 QUALITY ASSURANCE

- A. Manufacturer's Qualifications:
 - 1. Minimum five years of experience in production of equipment substantially similar to the specified equipment.
 - 2. Submit evidence of satisfactory operation of equipment similar to the specified equipment in at least three separate projects in accordance with the following requirements:
 - a. All equipment on the submitted installation list shall use the same design for critical components as specified for this project.
 - b. Multiple hot tapping and line stopping procedures for single project shall be considered as one installation requirement toward meeting the experience requirements.
- B. Fulfillment of the specified experience requirements shall be a condition of acceptance.
- C. The manufacturer of the hot tapping and line stopping devices shall assume complete responsibility for furnishing the specified equipment and shall have sole-source responsibilities for furnishing the complete installation assemblies and meeting the specified requirements of the procedure.

PART 2 PRODUCTS

2.01 TAPPING SLEEVE AND VALVE

- A. Tapping sleeves ductile-iron pipe:
 - 1. Tapping sleeve for ductile-iron pipe: Mueller 11-615 or equal.
 - 2. Pressure rating shall be at least 250 psi for piping 14 through 24 inches.
 - 3. Bolts: Stainless steel.
 - 4. Nozzle:
 - a. Diameter: 12 inches.
 - b. Flange: ASME 150-pound drilling, steel.
 - c. Test Port: 3/4-inch.
- B. Coating for tapping sleeves: Coat with epoxy per Section 09960 - High-Performance Coatings.
- C. Tapping valves: Provide permanent tapping valve to be connected to the tapping sleeve.
 - 1. Valve to conform to Section 15110 - Common Work Results for Valves and Section 15115 - Gate, Globe, and Angle Valves.
 - 2. Valve to be flanged, ductile iron, resilient seat, NRS, and designed for buried service with O-Ring stem packing and a stainless steel shaft.
 - 3. Equip valve with a valve box per Typical Detail P 022.

2.02 LINE STOP SYSTEM

- A. General Requirements:
 - 1. The line stop manufacturer must also perform the hot tapping process for the line stop.
- B. Design Requirements:
 - 1. The line stop must not slip or leak under the following conditions:
 - a. Maximum water velocity: 12 feet per second.
 - b. Rated back pressure 0 to 20 pounds per square inch.
 - 2. Line stop must be rated for a minimum of 100 pounds per square inch or higher.
- C. Plugging mechanism shall be advanced and retracted from the pipeline by means of hydraulic or mechanical actuator.
- D. When retracted, the plug head shall be housed in pressure-tight chamber between the actuator and tapping valve.
- E. Plug shall be capable of displacing accumulated deposits in interior of the pipe while advancing to its final position, have molded rubber or polyurethane sealing element around its perimeter, and supply workable seal with interior diameter when fully advanced.
- F. The equipment shall be folding-type assembly to allow for a reduced entry hole instead of a full size cut to retain the pipe stability.
- G. Installer shall provide means to address sealing issues caused by out of round and pipe imperfections.

PART 3 EXECUTION

3.01 GENERAL

- A. Upon providing the required advance notice to the Owner and receiving authorization, Contractor shall excavate the existing pipe at the proposed tap locations prior to ordering the tapping sleeve or saddle.
- B. Contractor shall fully expose the pipe within the limits of the proposed tapping sleeve or saddle and shall verify the type of existing pipe, the outside diameter, roundness, joint length, and condition of the pipe on which the tapping sleeve/saddle is to be installed.
- C. If any existing joint, fitting, tap, or other obstruction is located at the proposed tap location, Contractor shall coordinate with Engineer to determine the appropriate adjustment to the proposed tap location.
- D. Verification of pipe O.D. prior to installation: Verify O.D. prior to ordering materials.

- E. Where possible, CONTRACTOR to verify lining material of existing ductile iron pipe and repair lining after hot tap operations are complete with similar lining or equal.
- F. The CONTRACTOR shall demonstrate ability to accomplish the WORK by staging a "dry run" simulation of the procedure prior to initiation. To accomplish this, the CONTRACTOR shall walk through each step of the hot tap and line stop installation, and show the ENGINEER every component needed to install the hot tap and line stop, including but not limited to, tools and materials, to ensure that all the required components are on-site and in place prior to beginning the procedure.

3.02 INSTALLATION

- A. Operating conditions (e.g., flow rates, operating pressure, etc.) shall be confirmed with Owner or Engineer prior to ordering materials. Equipment and methods used to perform the tap shall be appropriate for the operating conditions.
- B. Install the sleeve and valve in strict accordance with the manufacturers' instructions and requirements of the Contract Documents. Tapping sleeve/saddle outlets shall be installed as indicated on the Drawings. Line stop may be installed vertical.

3.03 FIELD QUALITY CONTROL

- A. Perform a hydrostatic test of the tapping sleeve and valve assembly after installation of the tapping sleeve and valve, but prior to making the tap.
 - 1. Test to a pressure equal to 2 times the maximum operating pressure.
 - 2. Test duration: 4 hours.
- B. All bolts and nuts shall be kept clean and shall be handled carefully.
 - 1. Bolts and nuts shall be kept free of nicks and shall not be tossed or thrown.
 - 2. Bolts/nuts shall not be over-torqued.
 - 3. Stainless steel bolts and nuts shall be assembled with an anti-seize coating to prevent galling.

END OF SECTION



References for Bid – Water Main Line Stopping Services

Listed below are the references for Line Stops.

Date	Contractor	Line Stop Size	Job Location
October 26, 2020	Gentry Brothers Inc	24" Line Stop on a CMLC water line	Badillo and Grand Covina, CA Contact: Wayne 626-482-2225
October 8, 2019 - January 9, 2020	Doty Brothers	16" Line Stop on a CMLC water line	533 Via Del Monte Palos Verdes Estates, CA Contact: Tim Gochenouer Phone 562-519-1052
September 12, 2019 - October 16, 2019	Pacific Hydrotech Corporation	20" Line Stop on a CMLC water line	American Water 455 Maple Avenue Torrance, CA Contact: Eduardo 951-943-8803
July 17, 2019 - July 30, 2019	Tierra Contracting	24" Line Stop on Ductile Iron Pipe	100 South Mills Road Ventura, CA 93003 Contact: Jason 805-698-0719
September 2018	Mamco Inc	30"-24"-20" Line Stops on a CMLC water line	Gerald Desmond Bridge Project Long Beach, CA Contact: Rami 909-896-6923
May 2017	Orion Construction	30" and 24" Line Stops on a CMLC water line	MCTC Corridor Project Genesee Road and Regents Road San Diego, CA Contact: Jason 760-509-6211
June 2016	Flatiron West	24" Line Stops on a CMLC water line	I-5 Genesee Project San Diego, CA Contact: Jake 619-971-3311

If you have any further questions regarding the references, please let me know.

Sincerely,
Anthony Schneller
Koppl Pipeline Services, Inc.



KOPPL PIPELINE SERVICES, INC.

Hot Tapping • Line Stopping • Pipe Freezing • Valve Insertion

Basic Hot Tapping Procedure

Bolt-on Flanged or Threaded Connection

2020

BASIC HOT TAPPING PROCEDURE

BOLT-ON CONNECTION TO PIPELINE

This procedure is not intended as a substitute for training. Proper training and familiarization of necessary equipment is required to perform any safe hot tapping operation.

General

Proper information is crucial to a successful hot tapping operation. An on-site inspection of the work site is recommended to determine the correct method, equipment, fittings, and procedures required for performing the hot tap.

The process should be discussed with all persons involved, Contractors, owners, operators, maintenance or safety personnel, and anyone else who may be affected by the hot tapping operation should be contacted, if not present during the inspection. At this time, all pertinent information of the piping system should be obtained, such as:

- Type of pipe
- Diameter of Pipe
- Wall Thickness
- Operating Pressure
- Pressure Rating on System
- Type of Internal or External Coatings
- Line Contents

This information should be recorded and distributed for verification prior to the actual hot tapping operation. The jobsite location must also be considered and documented. This information may include:

- Pipe Location
- Method of Rigging Equipment
- Ingress and Egress to Jobsite
- Specific Site Requirements

Hot Tap Fitting Installation Procedure

There are several types of bolt-on fittings that are manufactured for performing a hot tap on a pipeline. This procedure will discuss general steps that must be taken to successfully complete a hot tap using one of the following four fitting designs:

- Service saddles, threaded outlet, double strap bands, cast with a single O-ring style seal, steel or stainless-steel straps, painted or epoxy coated.
- All stainless-steel tapping sleeve, threaded outlet, full encirclement rubber waffle sealing element, stainless steel bolts, nuts and washers.
- Fabricated steel or stainless-steel tapping sleeve, flanged outlet, single O-ring style seal, possibly painted or epoxy coated, steel or stainless-steel bolts, nuts and washers.
- All stainless-steel tapping sleeve, flanged outlet, full encirclement rubber waffle sealing element, stainless steel bolts, nuts and washers.

NOTE: The following instructions are for general use only. Even though the installation instructions for the above styles of hot tap fittings are similar, it is very important to refer to the manufacturer's installation instructions for each fitting. Some differences exist between manufacturers, especially on bolting sequences and torque ratings.

BASIC HOT TAPPING PROCEDURE

BOLT-ON CONNECTION TO PIPELINE

1. Inspect the flanged or threaded hot tap fitting to ensure no damage has occurred during shipment and storage. Check the pipeline outside diameter where the hot tap fitting will be installed. Check the size and range of the fitting to verify it is the correct fitting for that pipeline.
2. Prepare the pipeline for the bolt-on fitting installation. Only qualified personnel should perform the installation of the hot tap fitting.
 - Determine the location where the flanged or threaded hot tap fitting will be installed on the pipeline. Mark the pipeline.
 - Thoroughly clean the pipeline surface so that it is smooth and free of dirt, coatings, insulation, corrosion, or other debris that may impair the gasket seal.
 - Disassemble the hot tap fitting nuts, washers, and bolts and set them aside being careful to keep them clean and free from nicks.
 - Lubricate the pipeline and fitting gasket with a soapy water solution to affect a good seal.
3. Reassemble the hot tap fitting on the pipeline per manufacturer's instructions. Pay special attention to the manufacturer's recommended bolting sequence and torque levels for the bolts and nuts.
4. After installation is complete, perform a pressure test on the fitting and check for leaks. After the test is complete, recheck the torque levels of the bolts.

Installation procedures for hot tap fittings with mechanical joint end seals and hot tap fittings for steel cylinder pipe (PCCP, CMLC, etc.) are discussed in separate hot tap procedures. For a copy of these procedures, contact our corporate office at the phone number at the end of this document.

Hot Tap Valve Installation

The installation of the tapping valve is best done in the presence of Koppl Pipeline Services personnel. This is to ensure proper alignment and valve orientation.

1. Flanged Valve
 - Before installing the valve, confirm the valve flange rating and proper gasket material.
 - Bolt the valve to the nozzle using the requested bolting materials, (i.e. zinc coated, stainless steel, B-7, etc.). Once installed, open the valve to check for proper valve operation and that the bore of the valve is concentric with that of the nozzle. Also check for proper bore dimension on the valve.
 - Before the hot tap is made, start a hot tap checklist with the following dimensions:
 - Outer face of the valve to the crown of the pipe
 - Outer face of the valve to the nearest point on the valve gate in the closed position.
 - Number of turns to open or close the valve gate.
2. Threaded Valve
 - Apply thread sealing compound to the fitting and both ends of the nipple, then thread one end of the nipple into the fitting and tighten with a wrench.
 - Apply thread sealing compound to both ends of the valve, then thread one end onto the end of the nipple and tighten with a wrench.
 - Once installed, open the valve to check for proper valve operation. Also check for proper bore dimension on the valve.
 - Before the hot tap is made, start a hot tap checklist with the following dimensions:
 - Outer face of the valve to the crown of the pipe
 - Outer face of the valve to the nearest point on the valve gate in the closed position.
 - Number of turns to open or close the valve gate

BASIC HOT TAPPING PROCEDURE

BOLT-ON CONNECTION TO PIPELINE

Hot Tap Machine Installation

The installation of the hot tapping equipment shall be performed with the assistance of Koppl Pipeline Services personnel.

1. Before installing the hot tap machine, complete the hot tap checklist with the following information:
 - Confirm that the pressure rating of the tapping adapter on the machine is the same as the valve.
 - Check for proper pilot drill (tip, wire, length)
 - Check for proper shell cutter (tip, depth, diameter)
 - Confirm distance to complete cut and check boring bar travel
 - Confirm that boring bar is completely retracted prior to installation
2. Install the hot tapping machine onto tapping valve aligning the machine with the valve.

Pressure Test

Prior to beginning the tapping operation, a pressure test shall be performed to confirm the integrity of the fitting, the valve, and the hot tapping machine. This test will be performed with a maximum 80 psi air test or a maximum 200 psi hydrotest (per the pipeline owner's request).

1. Perform the following steps for a pressure test on a saddle with a test port
 - Close gate on valve
 - Remove test port from tapping saddle
 - Connect test equipment to test port
 - Initiate pressurization up the recommended or required test pressure
 - Check for leaks around saddle edges, the connection between the saddle and valve, the gate on the valve, and the valve packing. This is done using a soapy water solution.
 - Continue the pressure test for the required duration, periodically checking for leaks. Usually 10 – 60 minutes.
 - When integrity is confirmed, bleed off pressure and remove test equipment.
2. Perform the following steps for a pressure test on a saddle with no test port
 - Install tapping machine or test cap with test port
 - Connect test equipment to test port
 - Initiate pressurization up the recommended or required test pressure
 - Check for leaks around saddle edges, the connection between the saddle and valve, and the valve packing. This is done using a soapy water solution.
 - Continue the pressure test for the required duration, periodically checking for leaks. Usually 10 – 60 minutes.
 - When leak integrity is confirmed, bleed off pressure and remove test equipment.
3. If any leaks are found during the pressure test, the saddle and valve must be re-inspected and pressure tested again. It may be necessary to remove and reinstall the tapping saddle to obtain a positive pressure test.

BASIC HOT TAPPING PROCEDURE

BOLT-ON CONNECTION TO PIPELINE

Performing the Hot Tap

The hot tapping operation shall only be performed by Koppl Pipeline Services personnel.

1. Use the following steps to perform the hot tap
 - Confirm valve is in the open position
 - Confirm bleeder valve on tapping adapter is in the open position
 - Advance the boring bar until the tip of the pilot drill reaches the crown of the pipe. Log the number of turns required to complete this step on the hot tap checklist.
 - Reverse boring bar five turns
 - Engage power source
 - Feed boring bar five turns and begin pilot drill penetration
 - When pilot drill penetrates the pipe wall, vent the air out of the tapping adapter as needed and close the bleeder valve.
 - Continue to advance the boring bar until the cutter completely penetrates the pipe wall.
 - Disengage the power source and retract the boring bar, cutter, and pilot drill beyond the valve gate (confirm number of turns). The Hot Tap is finished.
2. Use the following steps to complete the Hot Tap
 - Close the valve gate
 - Open bleeder valve on tapping adapter to depressurize the machine. Proper disposal procedures should be discussed prior to venting the adapter.
 - Remove the tapping machine from the valve and confirm coupon retention
 - Remove coupon from the cutter and dispose of properly

Any techniques or methods listed in this procedure refer to equipment used by Koppl Pipeline Services. They may not necessarily apply to other equipment and should not be used as a substitute for other procedures.

Koppl Pipeline Services, Inc.
1228 Date Street
Montebello, CA 90640
Phone: 323-888-2211
Fax: 323-888-2232
www.koppl.com
CCL#844802
AZCL#ROCZ16719
NVCL#0060642



KOPPL PIPELINE SERVICES, INC.

Hot Tapping • Line Stopping • Pipe Freezing • Valve Inserting

Line Stop Procedure

Bolt-on Flanged or Threaded Connection

2020

LINE STOPPING PROCEDURE

BOLT-ON CONNECTION TO PIPELINE 2020

This procedure is not intended as a substitute for training. Proper training and familiarization of necessary equipment is required to perform any safe hot tapping or line stopping operation.

On-Site Inspection

1. Proper information is crucial to a successful line stopping operation. An on-site inspection of the work site is recommended to determine the correct method, equipment, fittings, and procedures required for performing the line stop. The process should be discussed with all persons involved, contractors, owners, operators, maintenance or safety personnel, and anyone else who may be affected by the line stopping operation should be contacted, if not present during the inspection. At this time, all pertinent information of the piping system should be obtained
 - Type of pipe
 - Inside and Outside Diameter of Pipe
 - Wall Thickness of steel can
 - Operating Pressure
 - Operating Temperature
 - Flange Rating on System
 - Type of Internal or External Coatings
 - Flow Rates
 - Line Contents
 - Type of work to be performed, time period line stop needs to be in line
2. This information should be recorded and distributed for verification prior to the actual line stopping operation. The jobsite location must also be considered and documented. This information may include
 - Pipe Location
 - Crane Requirements
 - Method of Rigging Equipment
 - Ingress and Egress to Jobsite
 - Specific Site Requirements

Line Stop Fitting Installation

1. There are several types of bolt-on fittings that are manufactured for performing a line stop on a pipeline. This procedure will discuss general steps that must be taken to successfully complete a line stop using one of the following fitting designs:
 - All stainless-steel sleeve, threaded outlet, full encirclement rubber waffle sealing element, stainless steel bolts, nuts and washers.
 - Fabricated steel or stainless-steel sleeve, flanged outlet, single O-ring style seal, possibly painted or epoxy coated, steel or stainless-steel bolts, nuts and washers.
 - All stainless-steel sleeve, flanged outlet, full encirclement rubber waffle sealing element, stainless steel bolts, nuts and washers.

NOTE: The following instructions are for general use only. Even though the installation instructions for the above styles of line stop fittings are similar, it is very important to refer to the manufacturer's installation instructions for each fitting. Some differences exist between manufacturers, especially on bolting sequences and torque ratings.

LINE STOPPING PROCEDURE

BOLT-ON CONNECTION TO PIPELINE 2020

2. Inspect the flanged or threaded line stop fitting to ensure no damage has occurred during shipment and storage. Check the pipeline outside diameter where the line stop fitting will be installed. Check the size and range of the fitting to verify it is the correct fitting for that pipeline.
3. Prepare the pipeline for the bolt-on fitting installation. Only Koppl Pipeline Services personnel should perform the installation of the line stop fitting.
 - Determine the location where the flanged or threaded line stop fitting will be installed on the pipeline. Mark the pipeline.
 - Thoroughly clean the pipeline surface so that it is smooth and free of dirt, coatings, insulation, corrosion, or other debris that may impair the gasket seal.
 - Disassemble the line stop fitting nuts, washers, and bolts and set them aside being careful to keep them clean and free from nicks.
 - Lubricate the pipeline and fitting gasket with a soapy water solution to affect a good seal.
4. Reassemble the line stop fitting on the pipeline per manufacturer's instructions. Pay special attention to the manufacturer's recommended bolting sequence and torque levels for the bolts and nuts.
5. After installation is complete, perform a pressure test on the fitting and check for leaks. After the test is complete, recheck the torque levels on the bolts.
6. Install the equalization fitting on the pipeline
 - Check for proper position of the fitting in relation to the line stop fitting
 - Confirm fitting ratings
 - Confirm proper fitting size
 - Use the manufacturer's instructions for the installation of the fitting onto the pipeline

Line Stop Temporary Isolation Valve Installation

1. The installation of the temporary isolation valve shall be performed with the assistance of Koppl Pipeline Services personnel. This is to ensure proper alignment and valve orientation.
 - Flanged Valve
 - Before installing the valve, confirm the valve flange rating and proper gasket material.
 - Bolt the valve to the nozzle using the required bolting materials. Once installed, open the valve to check for proper valve operation and that the bore of the valve is concentric with that of the nozzle. Also check for proper bore dimension on the valve.
 - Threaded Valve
 - Apply thread sealing compound to the fitting nipple and both ends of the threaded valve.
 - Thread one end of the valve onto the fitting nipple and tighten with a wrench. Be careful not to over-tighten the connection because the fitting nipple may become out of round to a point where the completion plug may not be able to be set. Once installed, open the valve to check for proper operation. Also check for proper bore dimension on the valve.

LINE STOPPING PROCEDURE

BOLT-ON CONNECTION TO PIPELINE 2020

2. Preset the completion plug

Presetting the completion plug is done to ensure proper alignment between the temporary isolation valve and the line stop fitting

- Attach the completion plug to the setting equipment
- Bolt the setting equipment to the temporary isolation valve
- Advance the completion plug through the valve to the set position in the fitting
- Make adjustments to the valve alignment if required
- Engage completion plug retention segments to confirm operation (count turns to set) or thread the completion plug fully into the line stop fitting
- Tighten bolting between valve and fitting
- Fully disengage completion plug retention segments or unthread the plug
- Fully retract completion plug into setting equipment (count the turns)
- Remove the setting equipment from temporary isolation valve

Hot Tap Machine Installation

The installation of the hot tap equipment shall be performed with the assistance of Koppl Pipeline Services personnel.

1. Before installing the hot tap machine, complete the hot tap checklist with the following information:
 - Outer face of the valve to the crown of the pipe
 - Outer face of the valve to the nearest point on the valve gate in the closed position.
 - Number of turns to open or close the valve gate
 - Confirm that the pressure rating of the tapping adapter on the machine is the same as the valve.
 - Check for proper pilot drill (tip, wire, length)
 - Check for proper shell cutter (tip, depth, diameter)
 - Confirm distance to complete cut and check boring bar travel
 - Confirm that boring bar is completely retracted prior to installation
2. Install the hot tapping machine onto the temporary isolation valve aligning the machine with the valve.

Pressure Test

Prior to beginning the tapping operation, a pressure test shall be performed to confirm the integrity of the fitting, the valve, and the hot tapping machine. This test will be performed with a maximum 80 psi air test or a maximum 200 psi hydro test (per the pipeline owner's request).

1. Perform the following steps for a pressure test on a line stop saddle with a test port
 - Close gate on valve
 - Remove test port from line stop saddle
 - Connect test equipment to test port
 - Initiate pressurization up the recommended or required test pressure
 - Check for leaks around saddle edges, the connection between the saddle and valve, the gate on the valve, and the valve packing. This is done using a soapy water solution.
 - Continue the pressure test for the required duration, periodically checking for leaks. Usually 10 – 60 minutes.
 - When integrity is confirmed, bleed off pressure and remove test equipment.

LINE STOPPING PROCEDURE

BOLT-ON CONNECTION TO PIPELINE 2020

2. Perform the following steps for a pressure test on a line stop saddle with no test port
 - Connect test equipment to test port on the tapping machine
 - Initiate pressurization up to the recommended or required test pressure
 - Check for leaks around saddle edges, the connection between the saddle and valve, the connection between valve and tapping machine, and the valve packing. This is done using a soapy water solution.
 - Continue the pressure test for the required duration, periodically checking for leaks. Usually 10 – 60 minutes.
 - When leak integrity is confirmed, bleed off pressure and remove test equipment.
3. If any leaks are found during the pressure test, the saddle and valve must be re-inspected and pressure tested again. It may be necessary to remove and reinstall the tapping saddle to obtain a positive pressure test.

Performing the Hot Tap

The hot tapping operation shall only be performed by Koppl Pipeline Services personnel.

1. Use the following steps to perform the hot tap
 - Confirm valve is in the open position
 - Confirm bleeder valve on tapping adapter is in the open position
 - Advance the boring bar until the tip of the pilot drill reaches the crown of the pipe. Log the number of turns required to complete this step on the hot tap checklist.
 - Reverse boring bar five turns
 - Engage power source
 - Feed boring bar five turns and begin pilot drill penetration
 - When pilot drill penetrates the pipe wall, vent the air out of the tapping adapter as needed and close the bleeder valve.
 - Continue to advance the boring bar until the cutter completely penetrates the pipe wall.
 - Disengage the power source and retract the boring bar, cutter, and pilot drill beyond the valve gate (confirm number of turns)
 - Close the valve gate
 - Open bleeder valve on tapping adapter and depressurize the machine. Proper disposal procedures should be discussed prior to venting the adapter.
 - Remove the tapping machine from the valve and confirm coupon retention
 - Remove coupon from the cutter
 - Verify the wall thickness by measuring the coupon, then dispose of the coupon properly
2. Check for contamination of the hot tapping machine
 - Any product that has contaminated the hot tap machine shall be removed on site and shall be the responsibility of the customer
 - Proper disposal of any hazardous waste generated by the cleaning of the machine shall also be handled by the customer

Line Stop Machine Installation

1. Installation of the line stop equipment shall be performed with the assistance of Koppl Pipeline Services personnel as follows
 - Confirm ID of the pipe and confirm that the OD of the sealing element is within tolerance

LINE STOPPING PROCEDURE

BOLT-ON CONNECTION TO PIPELINE 2020

- Attach the sealing element to the line stop head
 - Attach the line stop head to the line stop setting equipment
 - Install line stopping equipment on the line stop valve
 - Open line stop valve slowly, check for leaks. Purge air from the line stop housing
 - Confirm that the contents of the line are in a static state with no flow
 - Advise operations personnel that the line stop is ready to be installed
2. Setting the line stop head into the pipeline
- When operations gives permission, feed the line stop machine until measurements indicate the line stop head is seated in the proper position
 - Open the equalization valve to confirm the line stop element has produced a workable seal. If the seal is not adequate, the equipment must be removed to determine a cause
 - Once a seal is confirmed, authorization may be given to operations personnel to perform the necessary modifications to the piping system

Line Stopping Equipment Equalization

The equalization of the line stopping equipment shall be performed with the assistance of Koppl Pipeline Services personnel

- Attach temporary piping or a hose to connect the equalization valve to the bleeder valve on the line stop housing
- Open the equalization valve, then open the valve on the line stop housing slowly. Allow pressure on both sides of the line stop head to equalize
- The line stop head may now be safely removed from the pipeline

Line Stop Equipment Removal

No equipment shall be removed or adjusted prior to equalization

- Completely retract the line stop head from the pipe until fully inside the housing
- Confirm the line stop head is beyond the valve gate before closing the valve
- Close the temporary isolation valve
- Close the valves on both ends of the equalization hose and depressurize the hose. Remove the hose
- Open the bleeder valve on the housing to depressurize and drain the contents. Proper disposal procedures should be discussed prior to draining the housing
- When the housing is completely drained, remove the line stop machine

Setting the Completion Plug

Installation of the completion plug shall be performed by Koppl Pipeline Services personnel

- Prepare the completion plug setting machine by attaching the completion plug. Retract the completion plug fully inside the machine adapter
- Install the completion plug setting machine on the temporary isolation valve
- Open the valve and check for leaks
- Feed the completion plug through the valve and into the flange. Confirm with measurements that the plug is in its set position
- Fully engage the completion plug
- Detach the completion plug from the setting machine and retract the bar slightly

LINE STOPPING PROCEDURE
BOLT-ON CONNECTION TO PIPELINE 2020

- Open the bleeder on the setting tool to relieve the pressure and confirm that the completion plug is sealed
- Fully retract the bar on the completion plug setting machine
- Remove the setting machine from the temporary isolation valve
- Remove the temporary isolation valve
- Install the threaded cap or a gasket and a permanent blind flange using customer specified bolting (i.e. zinc coated, stainless steel, B-7, etc)
- The project is now complete

Any techniques or methods listed in this procedure refer to equipment used by Koppl Pipeline Services. They may not necessarily apply to other equipment and should not be used as a substitute for other procedures.

Koppl Pipeline Services, Inc.
1228 Date Street
Montebello, CA 90640
Phone: 323-888-2211
Fax: 323-888-2232
www.koppl.com
CCL#844802
AZCL#ROCZ16719
NVCL#0060642

12" X 24" BOLT ON SADDLE

KOPPL HOT TAP UNIT
M120-36 E/HD

EXIST. DIP
O.D. = 25.80"

24'

5.75"

14"

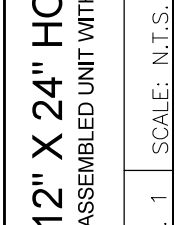
5'

PROFILE VIEW

Confidentiality:
Information contained on this drawing is confidential and may not be modified, copied, published, or otherwise distributed without the written consent of the company.

Confidentiality:
Information contained on this drawing is confidential and may not be modified, copied, published or used for any other purpose than was intended.

Confidentiality:
Information contained on this drawing is confidential and may not be modified, copied, published, or used for any other purpose than was intended.



12" X 24" HC
ASSEMBLED UNIT WITH

1	SCALE: N.T.S.
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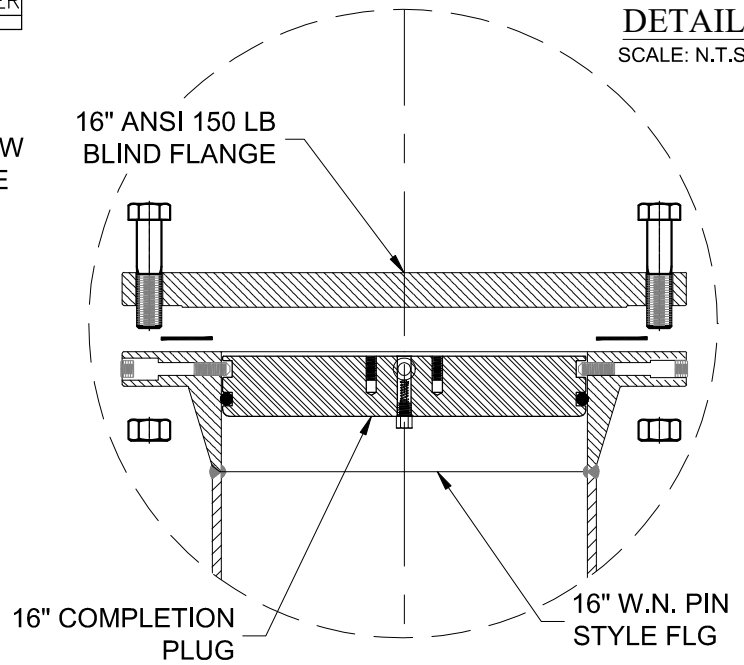
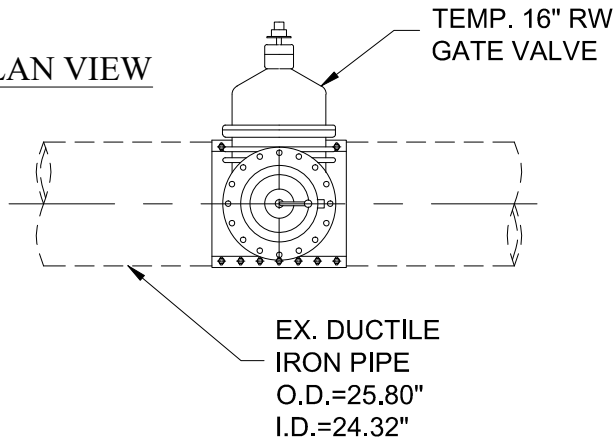
1	SCALE: N.T.S.
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	6/10/14	IBARRA	SCHNELLER
REVISIONS	DATE	BY	APPROVED

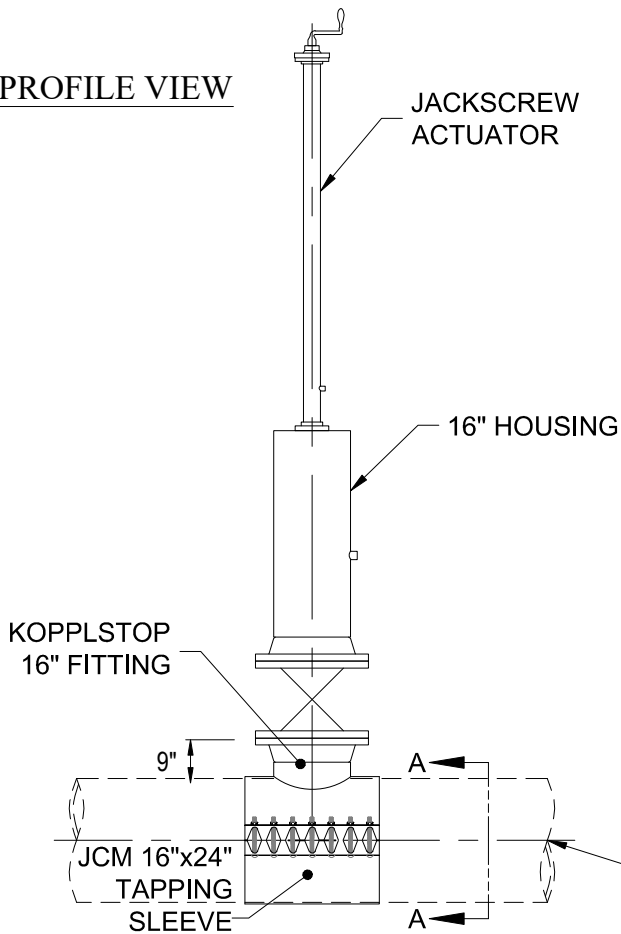
JACK SCREW SIZE	JACK SCREW SPAN	HOUSING SIZE	HOUSING LENGTH	RWGV LENGTH	FITTING LENGTH	L
KJS-072	82- $\frac{1}{4}$ "	16"	48"	16"	9"	12'-2"
KJS-084	100- $\frac{1}{4}$ "	16"	48"	16"	9"	13'-8"

DETAIL
SCALE: N.T.S.

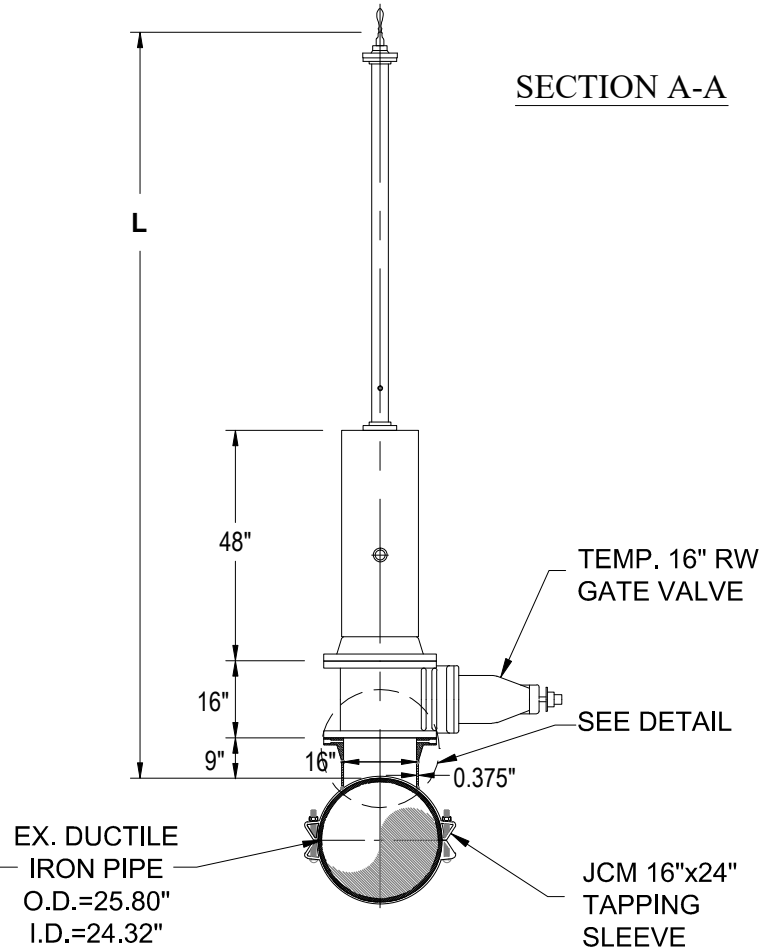
PLAN VIEW



PROFILE VIEW



SECTION A-A



Koppl Pipeline Services, Inc.

1228 Date Street
Montebello, CA 90640

(323) 888-2211 office
(323) 888-2232 fax

www.koppl.com

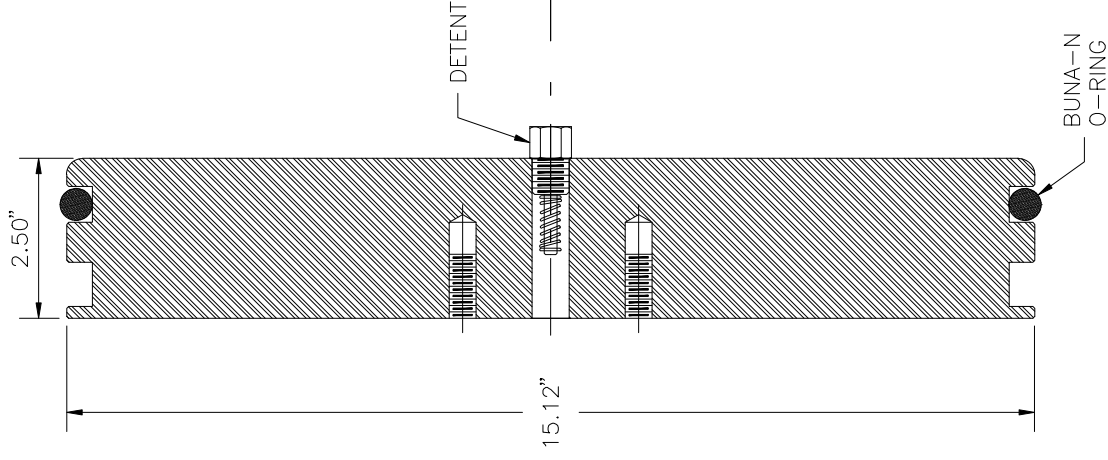
16" X 24" LINE STOP
FOLDING HEAD ASSEMBLED UNIT
FOR D.I. PIPE W/ RESILIENT WEDGE GATE VALVE

SCALE: N.T.S.

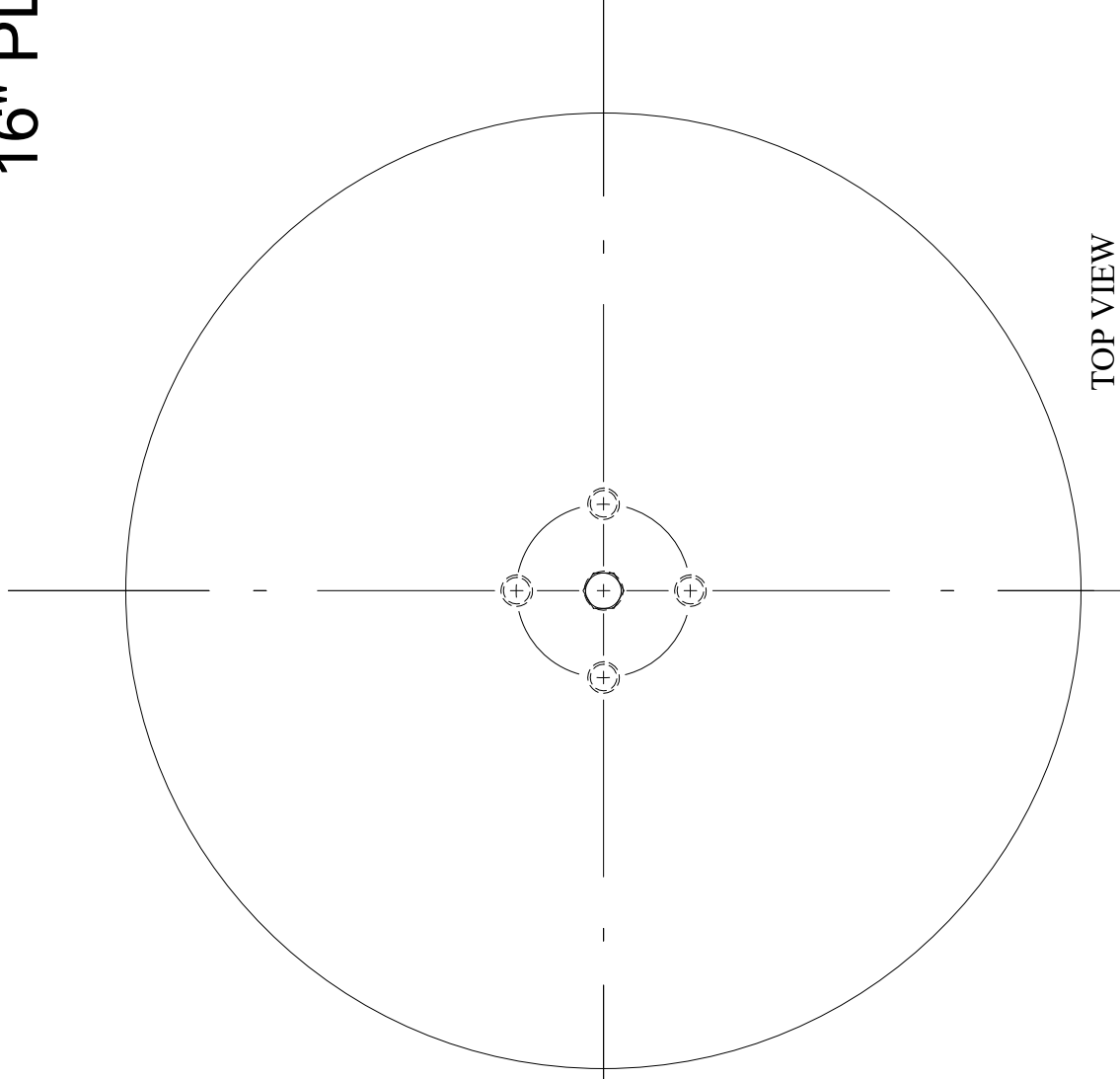
DWG: LS1624-DR

SECTION VIEW

16" PLUG



TOP VIEW



Koppl Pipeline Services, Inc.

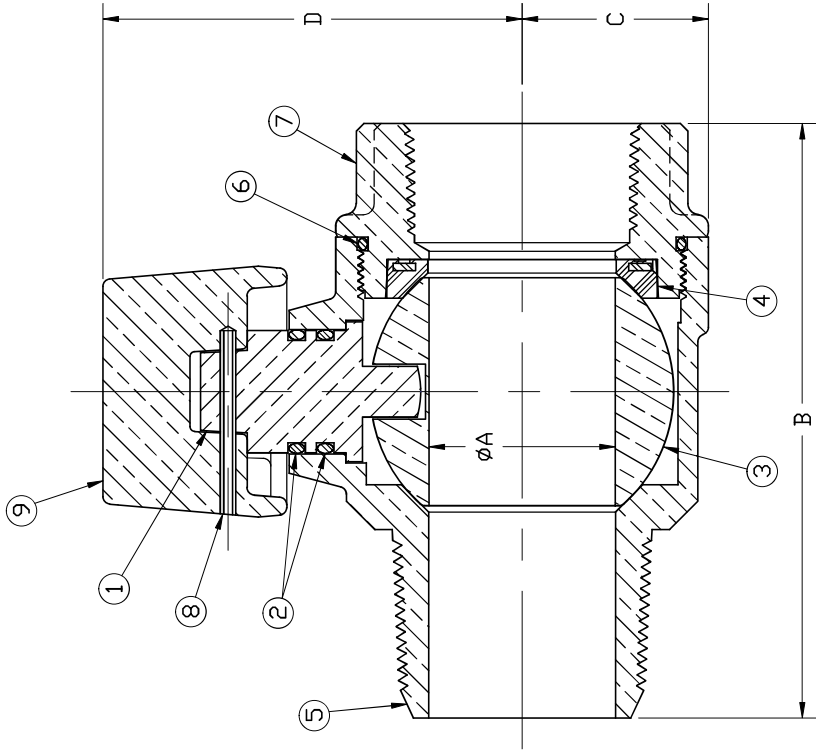
16-inch 150# Completion Plug
KOPPL PIN STYLE LINE STOP FITTING

REV. 3 SCALE: 1"=3" DWG # KSD-000055

1228 Date Street
Montebello, CA 90640
(323) 888-2211 office
(323) 888-2232 fax
www.koppl.com

REVISIONS	DATE	BY	APPROVED
remove dimension info.	11/19/13	IBARRA	SANCHEZ
adjust for larger set screw	11/15/13	IBARRA	DIAZ
	10/22/13	IBARRA	SCHNELLER

No.	DESCRIPTION	MATERIAL
1	STEM	LOW LEAD BRASS ALLOY
2	O-RING	EPDM D2000
3	BALL	LOW LEAD BRASS ALLOY
4	GASKET - SEAL	NITRILE D2000
5	BODY	CAST LOW LEAD BRASS ALLOY
6	O-RING	EPDM D2000
7	END PIECE	CAST LOW LEAD BRASS ALLOY
8	ROLL PIN	STAINLESS STEEL
9	CAP	CAST BRASS ASTM B62 ALLOY C83600



SIZE	φA	B	C	D
1-1/2"	1.50	4.78	1.50	3.37
2"	2.00	5.33	1.82	3.70

NOTE: ALL WETTED PARTS ARE LOW LEAD BRASS ALLOY

FILEPATH: PROD-ENG\MUELLER\ASSEMBLY\B2004510N

DRFTR	RWB	7/27/16	Muellet Co.	THIRD ANGLE PROJECTION
CHKR				
ENGR				
MGR				
UNLESS OTHERWISE NOTED			SURFACE PREPARATION: --- 000 RMS	
BREAK CORNER(S) --- 000			REF. No.	
SURFACE FINISH --- 000 RMS			SCALE	
TOLERANCE(S):			NONE	
LINEAR --- ± .000				
ANGULAR --- ± 0°00'				
STK No:				
MAT'L:				
REV	ECR	DESCRIPTION	DATE	BY
DESCR: BALL LLB CORPORATION VALVE				
DVG NUMBER B2004510N				
PER EST NO: DXXXX EST FIN WT = 0.0000 lbs				

DWG NUMBER
B2004510N
442-MRKTNG

TITLE TO AND OWNERSHIP OF THIS ENGINEERING
DATA REMAINS IN MUELLER CO. NO USE IS TO BE
MADE OF THIS DATA WITHOUT THE WRITTEN
AUTHORIZATION OF MUELLER CO. ASSENT IN THE PART
OF THE RECIPIENT TO THESE CONDITIONS IS PRESUMED



Installation Instructions

Model 412 Fabricated Carbon Steel Tapping Sleeve Flanged Outlet

Read instructions before starting installation*

Review of "Tricks of the Trade" on the reverse will assist with installation.

For purposes other than water, contact JCM Industries for application and product assistance.

1. Thoroughly clean pipe surface. Check the size and range of the tapping sleeve to verify correct size product. Check pipe surface where gasket will seat to make certain pipe is free of flaws, gouges and extreme irregularities.
2. Lubricate pipe and face of gasket with soap-water or gasket lubricating solution. **Do not** use grease or pipe lubricant. Oil based lubricants can prevent the gasket from adhering to pipe surface for water tight seal.
3. Position outlet half of body on pipe, making sure outlet is aligned with branch line to be connected. Never position so that rotation is required. Rotation can result in gasket dislocation.
4. Position back half of body and install bolts. (For fittings provided with stainless steel hardware, see reverse side.) **NOTE:** For fittings with hardware furnished with double washer sets, to assure ease of installation and obtain optimum bolt torque levels, install the double set of washers under each nut.

Tighten outside bolts first, working toward the center. Tighten bolts evenly. Alternate from one side of sleeve to the other. The gap between sleeve halves should be equal on both sides. JCM recommends the use of a torque wrench to ensure proper torque levels. Improper torque levels can result in leaking assembly or damage to the pipe wall.

Tighten bolts to the following torque levels:

Pipe sizes 6" - 12"	100 ft. lbs. of torque
Pipe sizes 14" & larger	125 ft. lbs. of torque
On Thin Wall, PVC (SDR21, 26), and Flexible Pipe	50 - 55 ft. lbs. minimum
HDPE SDR11, SDR17 - 6" - 12"	60 ft. lbs. minimum
HDPE SDR11, SDR17 - 14" and Larger	90 ft. lbs. minimum

**NOTE: For test and working pressure above 250 PSI bolts must be tightened to 125 - 150 ft. lbs. of torque.
(Contact JCM for proper application)**

5. Check inside of sleeve neck to make certain gasket is properly seated and not protruding where tapping cutter may damage it.
6. Install tapping valve. JCM recommends adherence to the AWWA M-44 Manual recommendations for proper valve installation, support and trenching. Improper support can result in undo stress.
7. Test assembly seals with water (per ANSI/AWWA C-223) using test port provided on sleeve or test connection on tapping machines. When testing the assembly against the pipe to pressures greater than the internal pressure of the host pipe, application should be treated with caution to prevent imploding or damaging the pipe wall due to thin wall, flexible or brittle conditions. Inspection and verification of the pipe integrity is the responsibility of the end user. For inquiries, contact JCM Industries, Inc.
8. When assured that all seals are tight and test is completed, re-check bolt torques after 15 minutes and proceed with the tapping operation. JCM Industries endorses the ANSI/AWWA C105/A21.5 and ASTM A674 for polyethylene encasement for corrosion protection.

Note:

Size on size tapping cutter must not be larger than recommended by pipe manufacturer. Tapping operation must not force the pipe away from the gasket seal.

INT412-0115

*Ensure fitting is suitable for application (confirm size, materials, pressure ratings, line content, meets local governing & association standards, etc.). Pipeline operation forces, including pressure fluctuations, thermal expansion/contraction, movement/shifting, etc. will influence the success of the application. Proper anchorage, restraint, harnessing, thrust blocks or other devices must be provided to prevent pipe movement (lateral, angular, axial) or pipe pullout from the bolt-on fitting. Inspection of the pipe integrity is the responsibility of the end user. JCM recommends the use of calibrated torque wrench. Failure to follow installation instructions will result in voided product warranty.

For application review or questions contact JCM Industries at 1-800-527-8482, 903-832-2581



Stainless Steel Fastener Management and Tips and Tricks Of The Trade For a Successful Installation

JCM Quality Fittings Equipped With 18-8 Stainless Steel Bolts and Nuts

When not properly handled it is the nature of stainless steel fasteners to gall and freeze (seize up). This is due to the inherent properties of the stainless material. Galling and freezing is often triggered by the presence of metal chips, burrs and grains of sand on the threads of the bolts and nuts. Extra care has been taken by JCM prior to assembly and packing of this fitting to assure a trouble-free installation.

1. The nuts and bolts are made from material of different hardness so that they have different strengths.
2. Nuts are coated with a special anti-seize coating. Additional lubricant may be needed. **A Molybdenum-Base lubricant is recommended.**
3. Each nut is assembled by hand to be sure that it went on the bolt freely.
4. The bolts and nuts are handled carefully to avoid damage to the threads.
5. The bolts and nuts are made to exacting specifications to assure that the correct material is used and that the thread form is correct.

Stainless hardware is especially susceptible to galling. JCM supplies specially coated nuts to eliminate the galling caused by over-torquing, but **the bolt threads must be kept clean, free from nicks and not pitched or thrown into the tool bucket during the installation process.** Use of the JCM 901 Master Wrench or JCM 905 Torque Wrench with Deep Socket is highly recommended. Use of pneumatic wrench for installation could cause hardware to seize and is not recommended.

Tricks of the Trade

Years of field experience, special applications and product testing have revealed many subtleties regarding application and installation of bolted fittings. For maximum performance under adverse conditions take advantage of the JCM "Tricks of the Trade."

Lubricate the pipe with soap-water or water. Oil based pipe lubricants produce a film between the gasket and pipe surface that is not water soluble and can interfere with the gasket/pipe water tight seal.

Do not rotate the sleeve on the pipe. Rotation of the sleeve on the pipe can result in the gasket being ripped from the groove and damaging the gasket beyond repair. Some manufacturers recommend rotation - JCM does not.

Tighten the bolts in the sequence provided in the instructions. Fittings are engineered to "load" the gasket in a certain fashion. Instructions provide the "sequence" of tightening bolts. Ensure the gap between sleeve halves is equal on both sides

Confirm bolt torques with a torque wrench. Proper compression of the gasket is critical to the success of the tapping sleeve installation. A torque wrench should be used to ensure recommended levels are achieved. Most field problems are directly related to lack of proper torque levels.

For water applications, proper valve installation, support and trenching per AWWA M-44. Improper assembly support and careless backfilling can sabotage an otherwise perfect installation.



Installation Instructions

Series 440 Fabricated Line Stop Sleeve

Types 2, 12, 3, 13, 5, 15

Read instructions before starting installation*

For purposes other than water, contact JCM Industries for application and product assistance.

1. Clean and scrape pipe. Remove any scale, pipe wrap, debris or dirt that may interfere with the complete sealing of the gasket. Inspect pipe for integrity, size, outside diameter and surface irregularities. Confirm the proper size and range of tapping sleeve. Inspect fitting to ensure all parts are included.

For fittings furnished with stainless steel hardware, see reverse for fastener management.

2. Lubricate the pipe and the fitting gasket with soapy water. Do not use oil base pipe lubricant.
3. Position outlet half of body on pipe, making sure outlet is aligned with branch line to be connected. Do Not position so that rotation is required. Rotation can result in gasket dislocation. NOTE: If pre-assembling valve to tapping sleeve before installing sleeve on pipe, for size on size sleeve install center bolts before installing the valve.
4. Position back half of body and install bolts. NOTE: For fittings with hardware furnished with double washer sets, to assure ease of installation and obtain optimum bolt torque levels, install the double set of washers under each nut.
5. Tighten outside bolts first, working toward the center. Tighten bolts evenly. Alternate from one side of sleeve to the other. The gap between sleeve halves should be equal on both sides. JCM recommends the use of a torque wrench to ensure proper torque levels. Improper torque levels can result in leaking assembly or damage to the pipe wall.

Tighten bolts to the following torque levels:

Pipe sizes 6" - 12" 100 ft. lbs. of torque

Pipe sizes 14" & larger 125 ft. lbs. of torque

NOTE: For test or working pressure above 250 PSI contact JCM for proper application. Bolts must be tightened to 125 - 150 ft. lbs. minimum of torque. Inspection of pipe integrity is responsibility of end user.

On Thin Wall, PVC (SDR21, 26), and Flexible Pipe 50 - 55 ft. lbs. minimum

HDPE SDR11, SDR17 - 6" - 12" 55 - 60 ft. lbs. minimum

HDPE SDR11, SDR17 - 14" and Larger 90 ft. lbs. minimum

6. Check inside of sleeve neck to make certain gasket is properly seated and not protruding where tapping cutter may damage it. Complete installation of fitting, return after approximately 15 minutes and confirm minimum bolt torque levels have been maintained. Test assembly seals using test plug provided on sleeve or test connection on tapping machine.
7. Install line stopping equipment and proceed with line stop operation.

For outlet sizes 16" and larger, prior to tapping, JCM recommends rebar and concrete encasement to support anticipated external loading (static and dynamic). The concrete shall be as close to the bottom of the outlet flange as possible (allowing for hardware clearance).

Alignment and support of the tapping sleeve is the responsibility of the end user, per best engineering practice, industry standard practice, or local code. For water applications: if applicable, test assembly seals with water (per ANSI/AWWA C-223). When testing the assembly against the pipe to pressures greater than the internal pressure of the host pipe, application should be treated with caution to prevent imploding or damaging the pipe wall due to thin wall, flexible or brittle conditions. No more than 10% above line pressure. Size on size tapping cutter must not be larger than recommended by pipe manufacturer. Tapping operation must not force the pipe away from the gasket seal. For inquiries, contact JCM Industries, Inc.

*Ensure fitting is suitable for application (confirm size, materials, pressure ratings, line content, meets local governing & association standards, etc.). Pipeline operation forces, including pressure fluctuations, thermal expansion/contraction, movement/shifting, etc. will influence the success of the application. Proper anchorage, restraint, harnessing, thrust blocks or other devices must be provided to prevent pipe movement (lateral, angular, axial) or pipe pullout from the bolt-on fitting. Inspection of the pipe integrity is the responsibility of the end user. JCM recommends the use of calibrated torque wrench. Failure to follow installation instructions will result in voided product warranty.

JCM Quality Fitting Equipped With 18-8 Stainless Steel Bolts and Nuts

When not properly handled it is the nature of stainless steel fasteners to gall and freeze (seize up). This is due to the inherent properties of the stainless material. Galling and freezing is often triggered by the presence of metal chips, burrs and grains of sand on the threads of the bolts and nuts. Extra care has been taken by JCM prior to assembly and packing of this fitting to assure a trouble-free installation.

1. The nuts and bolts are made from material of different hardness so that they have different strengths.
2. Nuts are coated with a special anti-seize coating. Additional lubricant may be needed. A Molybdenum-Base lubricant is recommended.
3. Each nut is assembled by hand to be sure that it went on the bolt freely.
4. The bolts and nuts are handled carefully to avoid damage to the threads.
5. The bolts and nuts are made to exacting specifications to assure that the correct material is used and that the thread form is correct.

Stainless hardware is especially susceptible to galling. JCM supplies specially coated nuts to eliminate the galling caused by over-torquing, but the bolt threads must be kept clean, free from nicks and not pitched or thrown into the tool bucket during the installation process. Use of a Torque Wrench with Deep Socket is highly recommended. Use of pneumatic wrench for installation could cause hardware to seize and is not recommended.

To ensure a successful, trouble free installation of this JCM Tapping Sleeve, the following Tips/Tricks are offered:

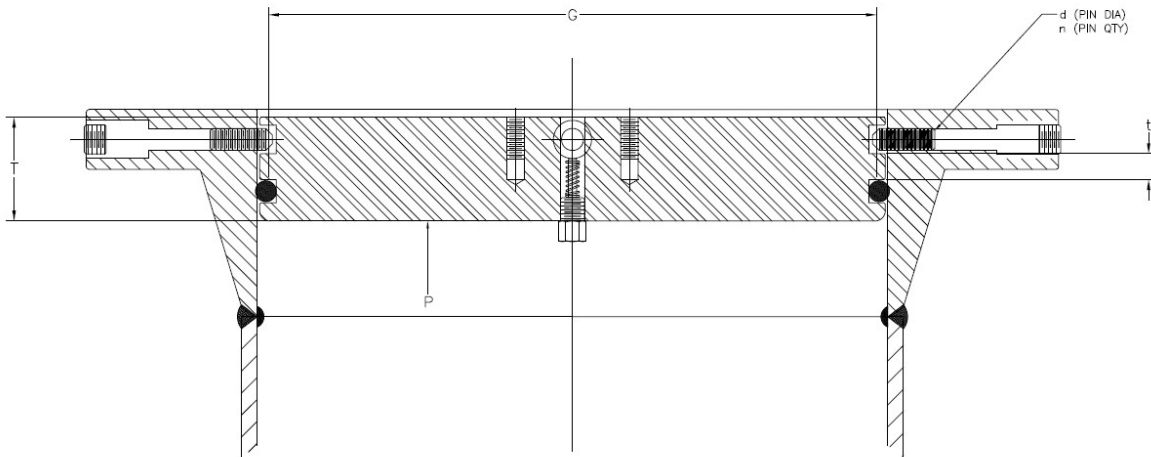
Lubricate the pipe with soap-water or water. Oil based pipe lubricants produce a film between the gasket and pipe surface that is not water soluble and can interfere with the gasket/pipe water tight seal.

Do not rotate the sleeve on the pipe. Rotation of the sleeve on the pipe can result in the gasket being ripped from the groove and damaging the gasket beyond repair. Some manufacturers recommend rotation - JCM does not.

Tighten the bolts in the sequence provided in the instructions. Fittings are engineered to "load" the gasket in a certain fashion. Instructions provide the "sequence" of tightening bolts. Ensure the gap between sleeve halves is equal on both sides

Confirm bolt torques with a torque wrench. Proper compression of the gasket is critical to the success of the tapping sleeve installation. A torque wrench should be used to ensure recommended levels are achieved. Most field problems are directly related to lack of proper torque levels.

16" Kopplstop Completion Plug Calculations



Design Pressure:	$P := 250$ psi	Pin Diameter:	$d := .625$ in
Gasket Diameter:	$G := 14.725$ in	Pin Quantity:	$n := 8$
Plate Thickness:	$T := 2.5$ in	Rib Thickness:	$t := 0.625$ in

Unstayed Flathead Thickness (ASME BPVC Sec VIII Div. 1, UG-34 Eq. 1)

Plate Allowable @ 100°F:	$S_b := 16600$ psi	(ASME BPVC Sec II, Part D)
Joint Efficiency:	$E := 1$	(Mechanically Restrained)
C-Factor:	$C := .30$	(Mechanically Restrained)

$$t_{min} := G \sqrt{\frac{C \cdot P}{S_b \cdot E}} = 0.99 \text{ in}$$

$$T > t_{min}$$

Check = "Plug meets minimum thickness."

Retaining Pin Check

Pin ultimate tensile strength:	$S_p := 229000$ psi	(ASTM F912 @ 47HRC UTS Conversion)
Pin Allowable in shear:	$S_\tau := 0.3 S_p = 68700$ psi	
Total pin area in shear:	$A_s = 2.45$ in²	

$$F := P \cdot \frac{\pi \cdot G^2}{4} = 42573.6 \text{ lbf}$$

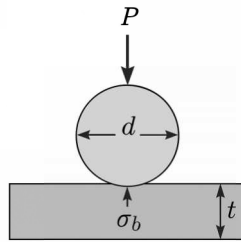
$$\sigma_{pin} := \frac{F}{A_s} = 17346.1 \text{ psi}$$

$$S_\tau > \sigma_{pin}$$

Check = "Pin meets minimum diameter."



Bearing Stress On Rib From Pin



$$P_n := \frac{F}{n} = 5321.7 \text{ } \textit{lb}f$$

$$\sigma_b := \frac{P_n}{d \cdot t} = 13623.6 \text{ } \textit{psi}$$

$$S_b > \sigma_b$$

Check = "Rib meets minimum thickness"



09/04/2025



P.O. Box 1220 • Nash • Texas • 75569

Phone: +1 (903) 832-2581 • www.jcmind.com

Date: 05/02/25

BODY, NECK, AND BOLTS : STRESS ANALYSIS

PRODUCT : 442-2580 X 16 ESS

RATED PRESSURE: 150 PSI

TEST PRESSURE: 225 PSI

BODY DESIGN	
BODY ID (in)	26.313
THICKNESS (in)	0.375
MATERIAL	A36
YIELD STRENGTH (psi)	36000
S.F.	4.43

NECK DESIGN	
NECK OD (in)	16.125
THICKNESS (in)	0.375
MATERIAL	A36
YIELD STRENGTH (psi)	36000
S.F.	7.44

FASTENERS DESIGN	
NO OF BOLTS	14
BOLT DIA (in)	0.75
MATERIAL	304 SS
YIELD STRENGTH (psi)	30000
SET TORQUE (ft-lb)	100
K	0.3
S.F.	1.06

BODY DESIGN

$$P = 2 \times S \times T / D$$

where,

P = Maximum Pressure Required to Burst Pipe

S = Yield Stress of Material

T = Material Thickness

D = Outside Diameter

$$P = 997.69 \text{ PSI}$$

Thus, S.F. = 997.69/225

$$\text{S.F.} = 4.43$$

NECK DESIGN

$$P = 2 \times S \times T / D$$

where,

P = Maximum Pressure Required to Burst Pipe

S = Yield Stress of Material

T = Material Thickness

D = Outside Diameter

$$P = 1674.4186 \text{ PSI}$$

Thus, S.F. = 1674.42/225

$$\text{S.F.} = 7.44$$

FASTENERS DESIGN

a. STRESS PER STUD, S(p)

Force on fitting during test,

$$F = (\text{Neck Area}) \times (\text{Test Pressure})$$

$$F = (\pi D^2/4) \times (\text{Test Pressure})$$

$$F = 58053.5 \text{ lbf}$$

where, D = Groove ID

$$= 18.13 \text{ in.}$$

Thus, force per stud = $F/(\text{No. of Bolts})$

$$\text{Force per stud} = 4146.6788 \text{ lbf}$$

Stress area per stud,

$$3/4''\text{-}10\text{unc} = 0.334 \text{ in}^2$$

$$\text{Thus, Stress per stud} = \frac{\text{Force per stud}}{\text{Stress area per stud}}$$

$$S(p) = 12415.206 \text{ lbf/in}^2$$

b. STRESS ON STUD BY TORQUE S(t)

Force applied by 100 ft-lb Torque,

$$F(t) = \frac{12 \times T}{K \times D}$$

$$\begin{aligned} \text{where, } T &= 100 \text{ ft-lb} \\ K &= 0.3 \\ D &= 0.75 \text{ in} \end{aligned}$$

$$F(t) = 5333.33 \text{ lbf}$$

Stress on stud by torque,

$$S(t) = \frac{F(t)}{\text{Stress area per stud}}$$

$$S(t) = 15968.06 \text{ lbf/in}^2$$

c. TOTAL STRESS ON STUDS

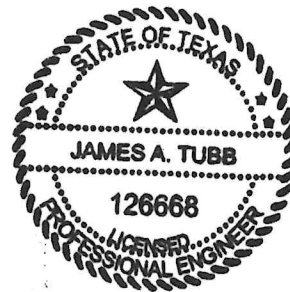
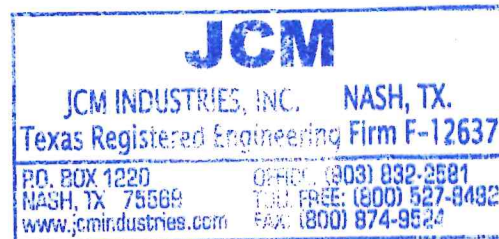
$$\text{Total Stress} = S(t) + S(p)$$

$$\text{Total Stress} = 28383.27 \text{ lbf/in}^2$$

Thus, S.F. on studs,

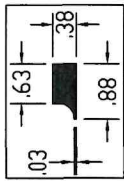
$$\text{S.F.} = 30000/28383.27$$

$$\text{S.F.} = 1.06$$

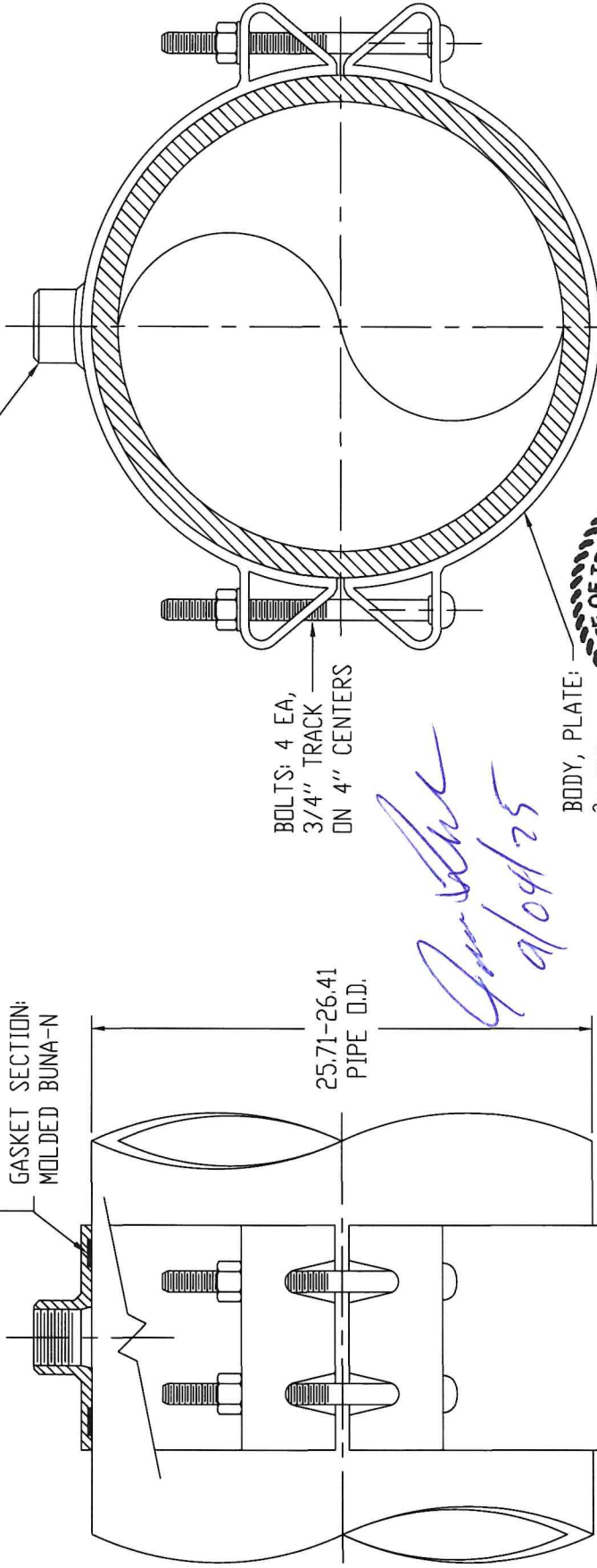


James A. Tubb
9/04/25

James Tubb, PE



GASKET SECTION:
MOLDED BUNA-N



BOLTS: 4 EA,
3/4" TRACK
ON 4" CENTERS

BODY, PLATE:
3/8" THK

SIDE/SECT

END VIEW

DESIGN CRITERIA:

CODE(S): AWVA C223
PRESSURE(PSI): 150
TEMPERATURE(*F): AMBIENT

NOTES:

1. FINISH: FUSION EPOXY COAT
2. FASTENERS: 304 SS



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JCM

JCM INDUSTRIES, INC. NASH, TX
OFFICE: (903) 832-9891
TOLL FREE: (800) 527-9482
P.O. BOX 1220 NASH, TX 75569
FAX: (800) 874-9524
www.jcmindustries.com

418-2580 X 14IP ESS

THREADED OUTLET BLOWDOWN SADDLE

UNLESS OTHERWISE SPECIFIED	DRAWN BY	CHECKED BY	APVD. BY	JWG. NO.	REV.
FRACTIONS (DECIMALS)	BTM	BTM	BTM	GEN-027056	
TOLERANCES TO BE	DATE	NO. SHEETS	SCALE	PART NO.	
± 1/8" .063 2°	10/06/23	1 of 1			

REVISIONS



P.O. Box 1220 • Nash • Texas • 75569

Phone: +1 (903) 832-2581 • www.jcmind.com

Date: 05/02/25

BODY, NECK, AND BOLTS : STRESS ANALYSIS

PRODUCT : 418-2580 X 14IP ESS

RATED PRESSURE: 150 PSI

TEST PRESSURE: 225 PSI

BODY DESIGN	
BODY ID (in)	26.438
THICKNESS (in)	0.375
MATERIAL	A36
YIELD STRENGTH (psi)	36000
S.F.	4.41

FASTENERS DESIGN	
NO OF BOLTS	4
BOLT DIA (in)	0.75
MATERIAL	304 SS
YIELD STRENGTH (psi)	30000
SET TORQUE (ft-lb)	100
K	0.3
S.F.	1.56

BODY DESIGN

$$P = 2 \times S \times T / D$$

where,

P = Maximum Pressure Required to Burst Pipe

S = Yield Stress of Material

T = Material Thickness

D = Outside Diameter

$$P = 993.10 \text{ PSI}$$

$$\text{Thus, S.F.} = 993.10 / 225$$

$$\text{S.F.} = 4.41$$

FASTENERS DESIGN

a. STRESS PER STUD, S(p)

Force on fitting during test,

$$F = (\text{Neck Area}) \times (\text{Test Pressure})$$

$$F = (\pi D^2 / 4) \times (\text{Test Pressure})$$

$$F = 4417.9 \text{ lbf}$$

where, D = Neck O.D.

$$= 5.00 \text{ in.}$$

Thus, force per stud = F / (No. of Bolts)

$$\text{Force per stud} = 1104.4662 \text{ lbf}$$

Stress area per stud,

$$3/4''\text{-}10\text{unc} = 0.334 \text{ in}^2$$

$$\text{Thus, Stress per stud} = \frac{\text{Force per stud}}{\text{Stress area per stud}}$$

$$S(p) = 3306.7849 \text{ lbf/in}^2$$

b. STRESS ON STUD BY TORQUE S(t)

Force applied by 100 ft-lb Torque,

$$F(t) = \frac{12 \times T}{K \times D}$$

$$\text{where, } T = 100 \text{ ft-lb}$$

$$K = 0.3$$

$$D = 0.75 \text{ in}$$

$$F(t) = 5333.33 \text{ lbf}$$

Stress on stud by torque,

$$S(t) = \frac{F(t)}{\text{Stress area per stud}}$$

$$S(t) = 15968.06 \text{ lbf/in}^2$$

c. TOTAL STRESS ON STUDS

$$\text{Total Stress} = S(t) + S(p)$$

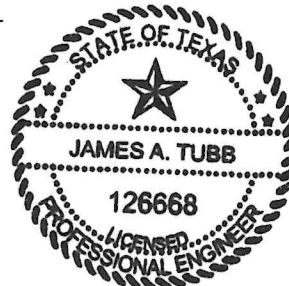
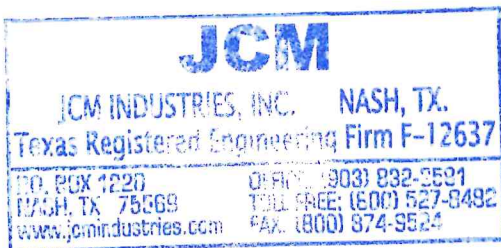
$$\text{Total Stress} = 19274.85 \text{ lbf/in}^2$$

Thus, S.F. on studs,

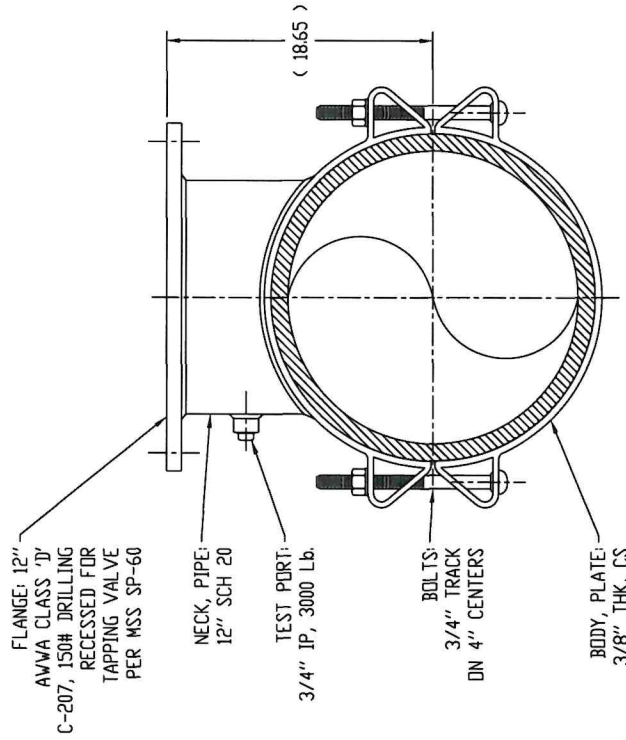
$$S.F. = 30000/19274.85$$

$$S.F. = 1.56$$

James Tubb, PE

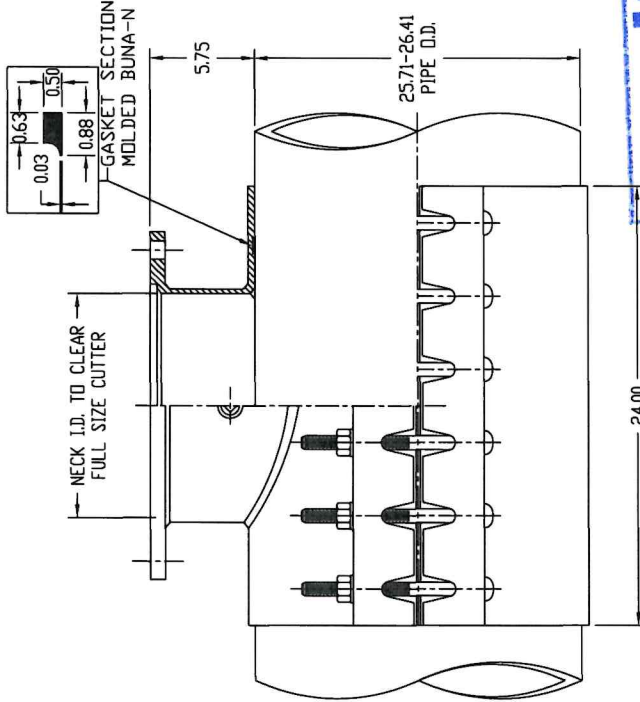


James Tubb
9/04/25

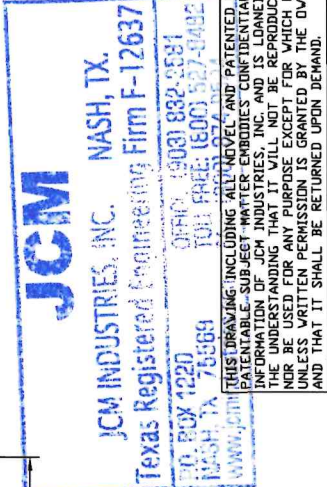


END VIEW

James A. Tubb
9/09/25



SECTION



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PATENTABLE SUBJECT MATTER EMBODIES CONFIDENTIAL
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UNLESS WRITTEN PERMISSION IS GRANTED BY THE OWNER
AND THAT IT SHALL BE RETURNED UPON DEMAND.

NOTES:

1. DESIGN & FAB. PER AWWA C-223
2. WORKING PRESSURE: 150 PSI
3. FINISH: FUSION EPOXY COAT
4. FASTENERS: 304 SS

JCM

JCM INDUSTRIES, INC. NASH, TX
P.O. BOX 1220
NASH, TX 75569
OFFICE (903) 832-2581
TOLL FREE (800) 527-8482
FAX (903) 832-2581
www.jcmindustries.com

412-2580 X 12 ESS

FLANGED OUTLET CS TAPPING SLEEVE

UNLESS OTHERWISE SPECIFIED	DRIVEN BY	CHECKED BY	APVD. BY	JWG. NO.	REV.
FRACTIONS (DECIMALS)	DATE	NO. SHEETS	SCALE	PART NO.	
± 1/4" .125 2"	1/29/19	1 of 1		GEN-017000	



P.O. Box 1220 • Nash • Texas • 75569

Phone: +1 (903) 832-2581 • www.jcmind.com

Date: 05/02/25

BODY, NECK, AND BOLTS : STRESS ANALYSIS

PRODUCT : 412-2580 X 12 ESS

RATED PRESSURE: 150 PSI

TEST PRESSURE: 225 PSI

BODY DESIGN	
BODY ID (in)	26.438
THICKNESS (in)	0.375
MATERIAL	A36
YIELD STRENGTH (psi)	36000
S.F.	4.41

NECK DESIGN	
NECK OD (in)	12.75
THICKNESS (in)	0.25
MATERIAL	A53B
YIELD STRENGTH (psi)	35000
S.F.	6.10

FASTENERS DESIGN	
NO OF BOLTS	12
BOLT DIA (in)	0.75
MATERIAL	304 SS
YIELD STRENGTH (psi)	30000
SET TORQUE (ft-lb)	100
K	0.3
S.F.	1.07

BODY DESIGN

$$P = 2 \times S \times T / D$$

where,

P = Maximum Pressure Required to Burst Pipe

S = Yield Stress of Material

T = Material Thickness

D = Outside Diameter

$$P = 993.10 \text{ PSI}$$

$$\text{Thus, S.F.} = 993.10 / 225$$

$$\text{S.F.} = 4.41$$

NECK DESIGN

$$P = 2 \times S \times T / D$$

where,

P = Maximum Pressure Required to Burst Pipe

S = Yield Stress of Material

T = Material Thickness

D = Outside Diameter

$$P = 1372.549 \text{ PSI}$$

$$\text{Thus, S.F.} = 1372.55 / 225$$

$$\text{S.F.} = 6.10$$

FASTENERS DESIGN

a. STRESS PER STUD, S(p)

Force on fitting during test,

$$F = (\text{Neck Area}) \times (\text{Test Pressure})$$

$$F = (\pi D^2/4) \times (\text{Test Pressure})$$

where, D = Groove ID

$$= 16.50 \text{ in.}$$

$$F = 48110.5 \text{ lbf}$$

Thus, force per stud = F/(No. of Bolts)

$$\text{Force per stud} = 4009.2122 \text{ lbf}$$

Stress area per stud,

$$3/4''\text{-}10\text{unc} = 0.334 \text{ in}^2$$

Thus, Stress per stud =

$$\frac{\text{Force per stud}}{\text{Stress area per stud}}$$

$$S(p) = 12003.629 \text{ lbf/in}^2$$

b. STRESS ON STUD BY TORQUE S(t)

Force applied by 100 ft-lb Torque,

$$F(t) = \frac{12 \times T}{K \times D}$$

where, $T = 100 \text{ ft-lb}$
 $K = 0.3$
 $D = 0.75 \text{ in}$

$$F(t) = 5333.33 \text{ lbf}$$

Stress on stud by torque,

$$S(t) = \frac{F(t)}{\text{Stress area per stud}}$$

$$S(t) = 15968.06 \text{ lbf/in}^2$$

c. TOTAL STRESS ON STUDS

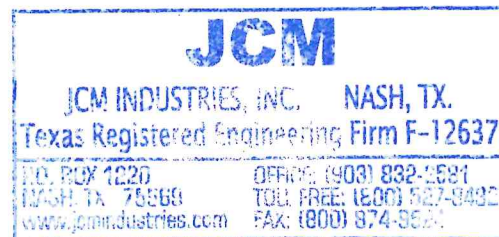
$$\text{Total Stress} = S(t) + S(p)$$

$$\text{Total Stress} = 27971.69 \text{ lbf/in}^2$$

Thus, S.F. on studs,

$$S.F. = 30000/27971.69$$

$$S.F. = 1.07$$



James A. Tubb
9/04/25

James Tubb, PE

2"-12" A-2362-E381 RESILIENT WEDGE GATE VALVE - FL. x FL.

- ☐ Catalog number – **A-2362-6** flanged ends
- ☐ Sizes – 2", 2-1/2", 3", 4", 6", 8", 10", 12"
- ☐ Meets or exceeds all applicable requirements of ANSI/AWWA C509 Standard, UL Listed, FM Approved, and certified to ANSI/NSF 61.†
- ☐ Flanged end dimensions and drilling comply with ANSI B16.1, class 125
- ☐ Iron body with nominal 10 mils MUELLER® Pro-Gard™ Fusion Epoxy Coated interior and exterior surfaces
- ☐ Epoxy coating meets or exceeds all applicable requirements of ANSI/AWWA C550 Standard and is certified to ANSI/NSF 61
- ☐ Iron wedge, symmetrical & fully encapsulated with molded rubber; no exposed iron
- ☐ Non-rising stem (NRS)
- ☐ Triple O-ring seal stuffing box (2 upper & 1 lower O-rings), with fourth O-ring serving as dirt seal††
- ☐ Handwheel (2" square wrench nut optional) – open left or open right
- ☐ 250 psig (1723 kPa) maximum working pressure, Max Temperature 120F
500 psig (3447 kPa) static test pressure
- ☐ 2-1/2"-12" sizes – UL Listed, FM Approved: 200 psig (1379 kPa)

† Approved for backflow prevention devices by USC (for 2-1/2" - 10" sizes)

††Dirt seal on 4"-12" valves



A-2362-6

Options

- ☐ Position indicators
- ☐ 2" square wrench nut

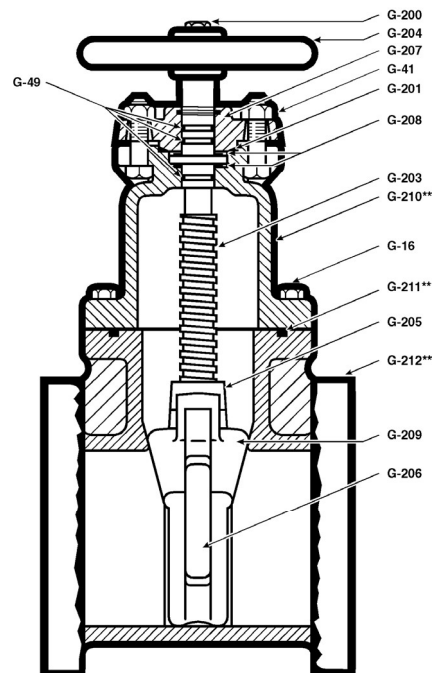
Resilient wedge gate valve parts

Catalog Part No.	Description	Material	Material standard
G-16	Bonnet Bolts & Nuts	Stainless Steel	Type 316
G-41	Stuffing Box Bolts & Nuts	Stainless Steel	Type 316
G-49	Stem O-rings (3)	EPDM	
G-200	Wrench Nut Cap Screw	Stainless Steel	Type 316
G-201	Stuffing Box Seal	EPDM	
G-202	Wrench Nut	Cast Iron	ASTM A126 CL.B
G-203	Stem	Everdur	ASTM B98
G-204	Hand Wheel (not shown)	Cast Iron	ASTM A126 CL.B
G-205	Stem Nut	Bronze	ASTM B62
G-206	Guide Cap Bearings	Celcon	
G-207	Stuffing Box with dirt seal	Cast iron Rubber	ASTM A126 CL.B --
G-208	Anti-friction Washers (2)	Celcon	
G-209	Wedge, Rubber Encapsulation	Cast Iron* EPDM	ASTM A126 CL.B --
G-210**	Bonnet	Ductile Iron	ASTM A536 Grade 65-45-12
G-211**	Bonnet O-ring	EPDM	
G-212**	Body	Ductile Iron	ASTM A536 Grade 65-45-12

* Fully encapsulated in molded rubber with no iron exposed

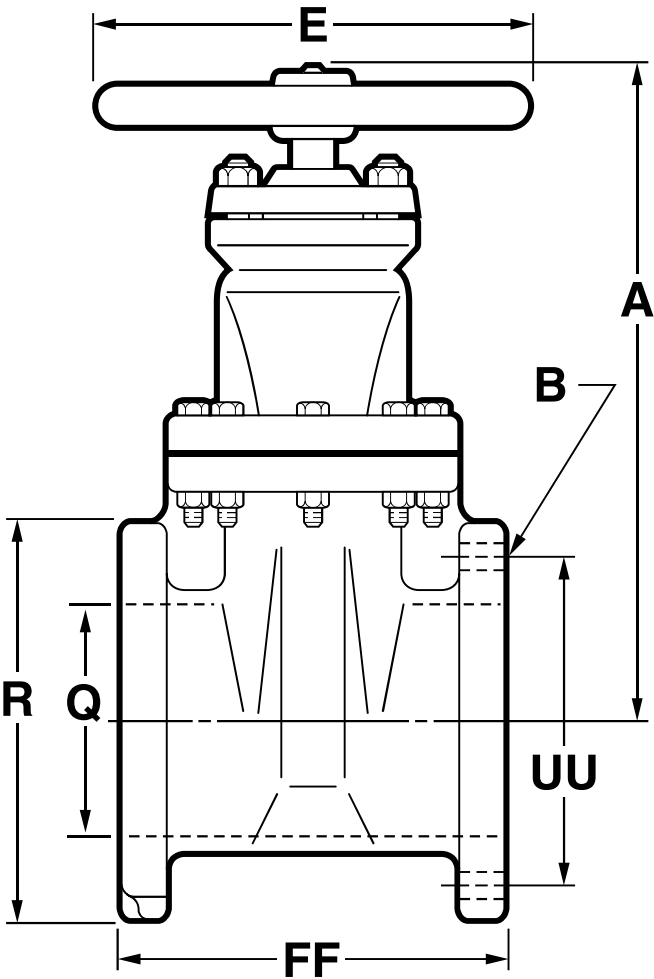
** Previous to 1999 these parts on 4"-12" valves were designed with a gasket instead of an O-ring and with additional bolts (2"-3" sizes retain neoprene gasket design affecting these parts). Confirm the type of seal when ordering a replacement gasket or O-ring.

††Dirt seal on 4"-12" valves



2"-12" A-2362-E381 RESILIENT WEDGE GATE VALVE - FL. x FL.

Mueller Co.



Dimensions

Dimension*	Nominal size							
	2"	2-1/2"	3"	4"	6"	8"	10"	12"
A	9.88	12.38	12.38	14.19	18.00	21.50	25.50	28.62
E	6.00	6.00	8.00	11.00	13.00	14.00	16.00	16.00
R	6.00	7.00	7.50	9.00	11.00	13.50	16.00	19.00
FF	7.00	7.50	8.00	9.00	10.50	11.50	13.00	14.00
Q (bore)	2.30	2.80	3.30	4.30	6.30	8.30	10.30	12.30
UU (bolt circle diameter)	4.75	5.50	6.00	7.50	9.50	11.75	14.25	17.00
B (number and size of holes)	4--.75	4--.75	4--.75	8--.75	8--.88	8--.88	12--1.00	12--1.00
Turns to open	8	11	11	14	20.5	26.5	33	38.5
Weight*	37	71	73	96	154	250	400	500

*All dimensions are in inches. All weights are in pounds and are approximate.

4"-12" A-2362-E381 RESILIENT WEDGE GATE VALVES - M.J. x FL.

- ☐ Catalog number–
A-2362-16 mechanical joint x flanged ends (with accessories unassembled)
A-2362-19 mechanical joint x flanged ends (less accessories)
- ☐ Sizes – 4", 6", 8", 10", 12"
- ☐ Meets or exceeds all applicable requirements of ANSI/AWWA C509 Standard, UL Listed, FM Approved, and certified to ANSI/NSF 61.
- ☐ Flanged end dimensions and drilling comply with ANSI B16.1, class 125
- ☐ Mechanical joint end complies with ANSI/AWWA C111 Standard
- ☐ Iron body with nominal 10 mils MUELLER® Pro-Gard™ Fusion Epoxy Coated interior and exterior surfaces
- ☐ Epoxy coating meets or exceeds all applicable requirements of ANSI/AWWA C550 Standard and is certified to ANSI/NSF 61
- ☐ Iron wedge, symmetrical & fully encapsulated with molded rubber; no exposed iron
- ☐ Non-rising stem (NRS)
- ☐ Triple O-ring seal stuffing box (2 upper & 1 lower O-rings), with fourth O-ring serving as dirt seal
- ☐ 2" square wrench nut (optional handwheel available) – open left or open right
- ☐ 250 psig (1723 kPa) maximum working pressure, 500 psig (3447 kPa) static test pressure
- ☐ UL Listed, FM Approved – 200 psig (1379 kPa)



A-2362-16

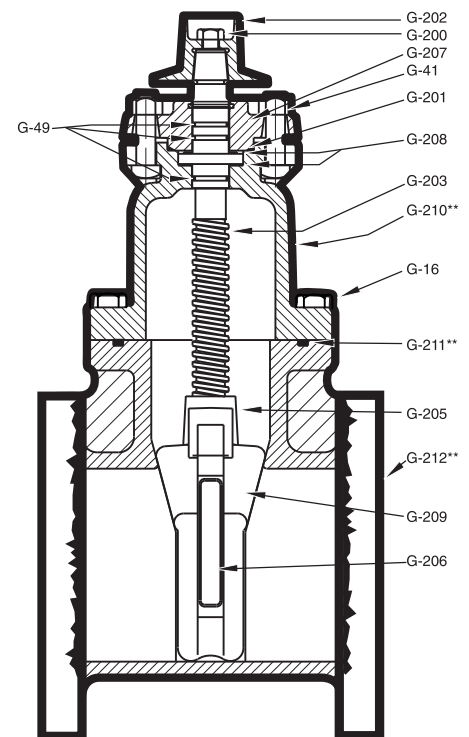
M.J. accessories
shipped unassembled

Options

- ☐ Position indicators
- ☐ Handwheel

Resilient wedge gate valve parts

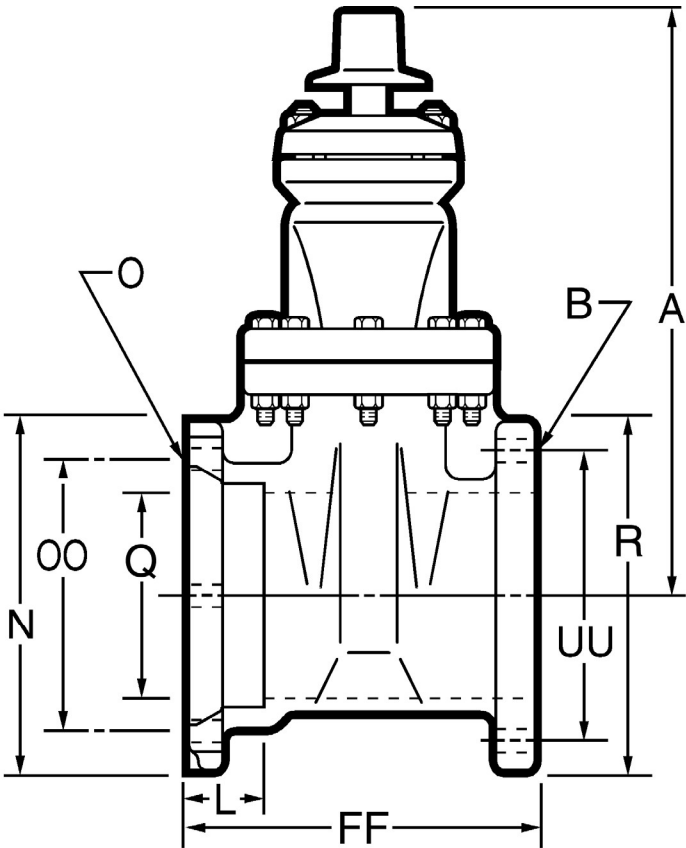
Catalog Part No.	Description	Material	Material standard
G-16	Bonnet Bolts & Nuts	Stainless Steel	Type 316
G-41	Stuffing Box Bolts & Nuts	Stainless Steel	Type 316
G-49	Stem O-rings (3)	EPDM	
G-200	Wrench Nut Cap Screw	Stainless Steel	Type 316
G-201	Stuffing Box Seal	EPDM	
G-202	Wrench Nut	Cast Iron	ASTM A126 CL.B
G-203	Stem	Everdur	ASTM B98
G-204	Hand Wheel (not shown)	Cast Iron	ASTM A126 CL.B
G-205	Stem Nut	Bronze	ASTM B62
G-206	Guide Cap Bearings	Celcon	
G-207	Stuffing Box with dirt seal	Cast iron Rubber	ASTM A126 CL.B --
G-208	Anti-friction Washers (2)	Celcon	
G-209	Wedge, Rubber Encapsulation	Cast Iron* EPDM	ASTM A126 CL.B --
G-210**	Bonnet	Ductile Iron	ASTM A536 Grade 65-45-12
G-211**	Bonnet O-ring	EPDM	
G-212**	Body	Ductile Iron	ASTM A536 Grade 65-45-12



* Fully encapsulated in molded rubber with no iron exposed

4"-12" A-2362-E381 RESILIENT WEDGE GATE VALVE - M.J. x FL.

Mueller Co.



Dimensions

Dimension*	Nominal size				
	4"	6"	8"	10"	12"
A	14.19	18.00	21.50	25.50	28.62
L	2.50	2.50	2.50	2.50	2.50
N	9.12	11.12	13.37	15.62	17.88
O (number and size of holes for MJ)	4--.88	6--.88	6--.88	8--.88	8--.88
Q (bore)	4.30	6.30	8.30	10.30	12.30
OO (bolt circle diameter for MJ)	7.50	9.50	11.75	14.00	16.25
R	9.00	11.00	13.50	16.00	19.00
UU (bolt circle diameter for FL)	7.50	9.50	11.75	14.25	17.00
FF	9.50	11.00	12.00	13.88	14.44
B (number and size of holes for FL)	8--.75	8--.88	8--.88	12--1.00	12--1.00
Turns to open	14	20.5	26.5	33	38.5
Weight*	115	168	275	400	570

* All dimensions are in inches. All weights include accessories are in pounds and are approximate.

2"-12" A-2362-E381 RESILIENT WEDGE GATE VALVES - M.J. x M.J.

- ☐ Catalog number—
A-2362-20 Mechanical joint ends (with accessories unassembled)
A-2362-23 Mechanical joint ends (less accessories)
A-2362-25 Mechanical joint ends (with transition gaskets accessories unassembled)
- ☐ Sizes – 2", 3", 4", 6", 8", 10", 12"
- ☐ Meets or exceeds all applicable requirements of ANSI/AWWA C509 Standard, UL Listed, FM Approved, and certified to ANSI/NSF 61.
- ☐ Standard mechanical joint ends comply with ANSI/AWWA C111
- ☐ Iron body with nominal 10 mils MUELLER® Pro-Gard™ Fusion Epoxy Coated interior and exterior surfaces
- ☐ Epoxy coating meets or exceeds all applicable requirements of ANSI/AWWA C550 Standard and is certified to ANSI/NSF 61
- ☐ Iron wedge, symmetrical & fully encapsulated with molded rubber; no exposed iron
- ☐ Non-rising stem (NRS)
- ☐ Triple O-ring seal stuffing box (2 upper & 1 lower O-rings), with fourth O-ring serving as dirt seal††
- ☐ 2" square wrench nut (optional handwheel available)—open left or open right
- ☐ 250 psig (1723 kPa) maximum working pressure, 500 psig (3447 kPa) static test pressure
- ☐ 3"-12" sizes – UL Listed, FM Approved: 200 psig (1379 kPa)

††Dirt seal on 4"-12" valves



A-2362-20 M.J. accessories shipped unassembled

Options

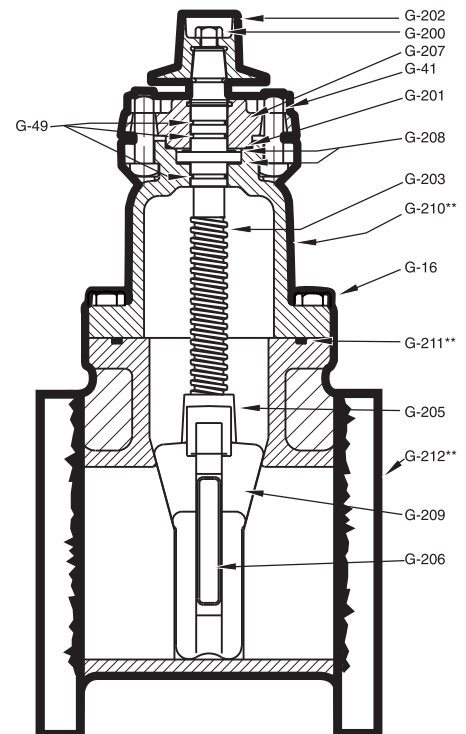
- ☐ Position indicators
- ☐ Handwheel

Resilient wedge gate valve parts

Catalog Part No.	Description	Material	Material standard
G-16	Bonnet Bolts & Nuts	Stainless Steel	Type 316
G-41	Stuffing Box Bolts & Nuts	Stainless Steel	Type 316
G-49	Stem O-rings (3)	EPDM	
G-200	Wrench Nut Cap Screw	Stainless Steel	Type 316
G-201	Stuffing Box Seal	EPDM	
G-202	Wrench Nut	Cast Iron	ASTM A126 CL.B
G-203	Stem	Everdur	ASTM B98
G-204	Hand Wheel (not shown)	Cast Iron	ASTM A126 CL.B
G-205	Stem Nut	Bronze	ASTM B62
G-206	Guide Cap Bearings	Celcon	
G-207	Stuffing Box with dirt seal	Cast iron Rubber	ASTM A126 CL.B --
G-208	Anti-friction Washers (2)	Celcon	
G-209	Wedge, Rubber Encapsulation	Cast Iron* EPDM	ASTM A126 CL.B --
G-210**	Bonnet	Ductile Iron	ASTM A536 Grade 65-45-12
G-211**	Bonnet O-ring	EPDM	
G-212**	Body	Ductile Iron	ASTM A536 Grade 65-45-12

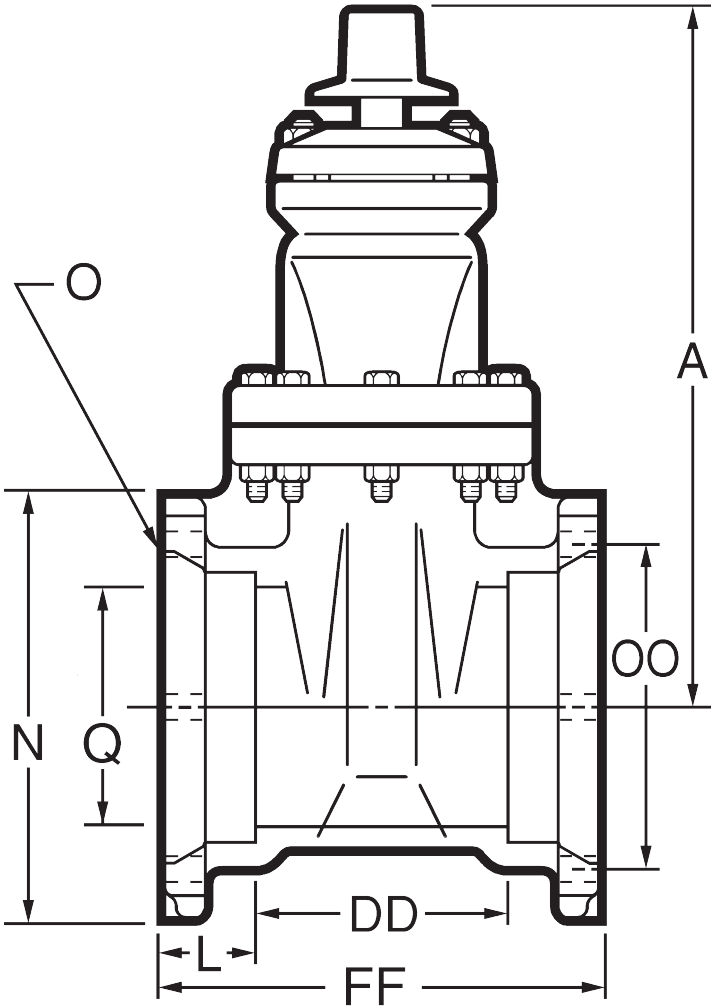
* Fully encapsulated in molded rubber with no iron exposed

††Dirt seal on 4"-12" valves



2"-12" A-2362-E381 RESILIENT WEDGE
GATE VALVE - M.J. x M.J.

Mueller Co.



Dimensions

Dimension*	Nominal size						
	2"	3"	4"	6"	8"	10"	12"
A	9.88	12.38	14.19	18.00	21.50	25.50	28.62
FF	8.50	9.00	10.00	11.50	12.50	14.75	14.88
L	2.50	2.50	2.50	2.50	2.50	2.50	2.50
N	4.62	7.50	9.12	11.12	13.37	15.62	17.88
O (number and size of holes)	4--.88	4--.88	4--.88	6--.88	6--.88	8--.88	8--.88
DD	3.50	4.00	5.00	6.50	7.50	9.75	9.88
Q (bore)	2.30	3.30	4.30	6.30	8.30	10.30	12.30
OO (bolt circle diameter)	5.00	6.19	7.50	9.50	11.75	14.00	16.25
Turns to open	8	11	14	20.5	26.5	33	38.5
Weight*	40	83	120	186	280	405	540

*All dimensions are in inches. All weights include accessories are in pounds and are approximate.

4"-12" A-2362-E381 RESILIENT WEDGE GATE VALVES - SL. x SL.

- ☐ Catalog number–
A-2362-38 Slip-on† ends with gasket
A-2362-40 Slip-on† ends less gasket
- ☐ Sizes – 4", 6", 8", 10", 12"
- ☐ Meets or exceeds all applicable requirements of ANSI/AWWA C509 Standard, UL Listed, FM Approved, and certified to ANSI/NSF 61.
- ☐ Slip-on ends fit plain end of classes 150, 200 and 250 cast iron; ductile iron and classes 150 and 200 cast iron O.D. PVC
- ☐ Iron body with nominal 10 mils MUELLER® Pro-Gard™ Fusion Epoxy Coated interior and exterior surfaces
- ☐ Epoxy coating meets or exceeds all applicable requirements of ANSI/AWWA C550 Standard and is certified to ANSI/NSF 61
- ☐ Iron wedge, symmetrical & fully encapsulated with molded rubber; no exposed iron
- ☐ Non-rising stem (NRS)
- ☐ Triple O-ring seal stuffing box (2 upper & 1 lower O-ring), with fourth O-ring serving as dirt seal
- ☐ 2" square wrench nut (optional handwheel available) – open left or open right
- ☐ 250 psig (1723 kPa) maximum working pressure, 500 psig (3447 kPa) static test pressure
- ☐ UL Listed, FM Approved: 200 psig (1379 kPa)



A-2362-38

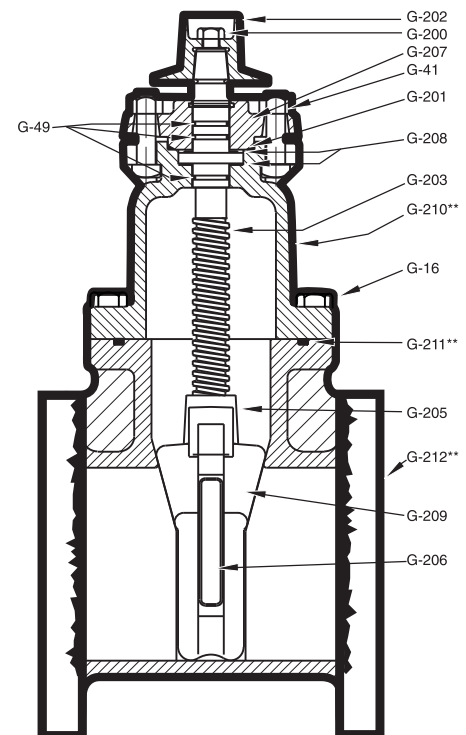
†Design and dimensions of the joint are manufactured under license of U.S. Pipe and Foundry Company.

Options

- ☐ Position indicators
- ☐ Handwheel

Resilient wedge gate valve parts

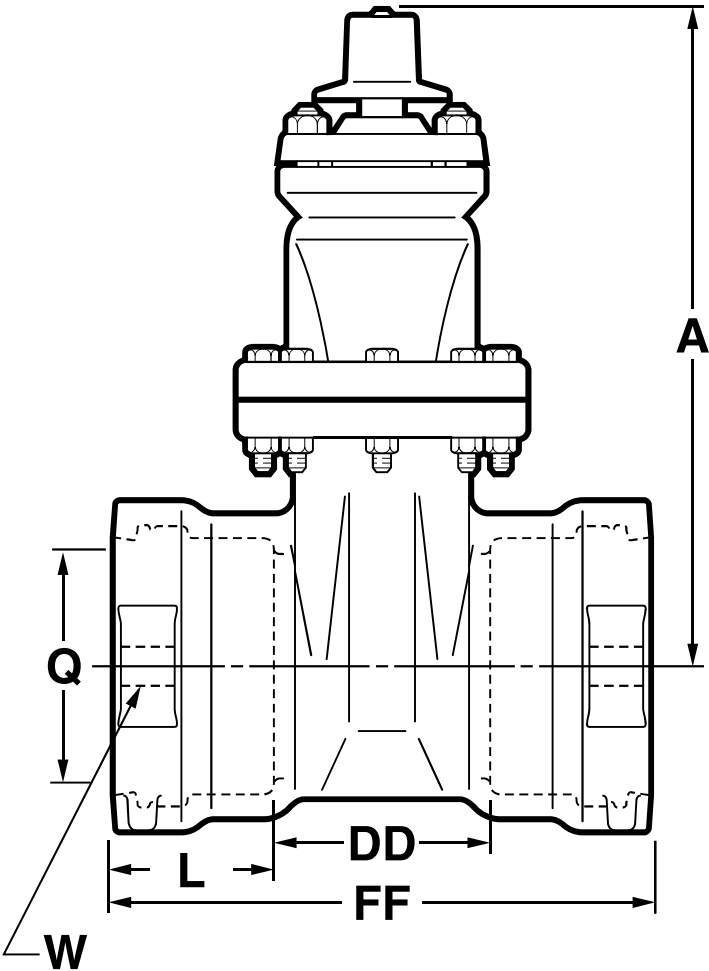
Catalog Part No.	Description	Material	Material standard
G-16	Bonnet Bolts & Nuts	Stainless Steel	Type 316
G-41	Stuffing Box Bolts & Nuts	Stainless Steel	Type 316
G-49	Stem O-rings (3)	EPDM	
G-200	Wrench Nut Cap Screw	Stainless Steel	Type 316
G-201	Stuffing Box Seal	EPDM	
G-202	Wrench Nut	Cast Iron	ASTM A126 CL.B
G-203	Stem	Everdur	ASTM B98
G-204	Hand Wheel (not shown)	Cast Iron	ASTM A126 CL.B
G-205	Stem Nut	Bronze	ASTM B62
G-206	Guide Cap Bearings	Celcon	
G-207	Stuffing Box with dirt seal	Cast iron Rubber	ASTM A126 CL.B --
G-208	Anti-friction Washers (2)	Celcon	
G-209	Wedge, Rubber Encapsulation	Cast Iron* EPDM	ASTM A126 CL.B --
G-210**	Bonnet	Ductile Iron	ASTM A536 Grade 65-45-12
G-211**	Bonnet O-ring	EPDM	
G-212**	Body	Ductile Iron	ASTM A536 Grade 65-45-12



* Fully encapsulated in molded rubber with no iron exposed

4"-12" A-2362-E381 RESILIENT WEDGE GATE VALVE - SL. x SL.

Mueller Co.



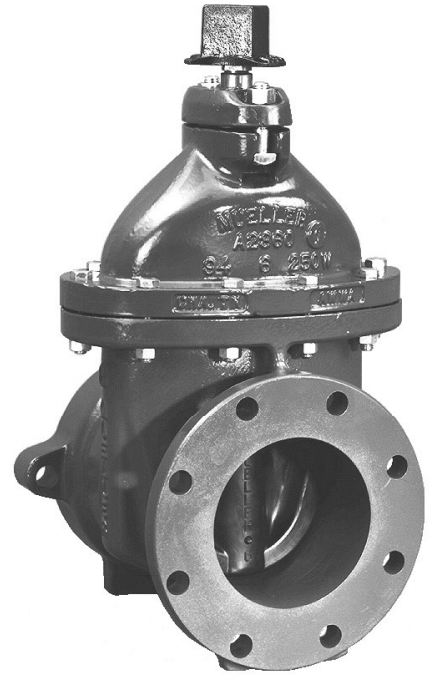
Dimensions

Dimension*	Nominal size				
	4"	6"	8"	10"	12"
A	14.19	18.00	21.50	25.50	28.62
Q (bore)	4.30	6.30	8.30	10.30	12.30
W (distance between strapping lug holes across face of valve ends)	8.62	10.62	13.12	16.38	18.12
FF	12.81	14.81	17.88	18.43	18.62
L	4.12	4.38	5.62	5.62	5.62
DD	4.57	6.05	6.64	7.19	7.38
Turns to open	14	20.5	26.5	33	38.5
Weight*	90	165	260	400	565

*All dimensions are in inches. All weights are in pounds and are approximate.

4"-12" A-2362-E381 RESILIENT WEDGE GATE VALVES - SL. x FL.

- ☐ Catalog number—
A-2362-41 Slip-on† end (with gasket) x flange end
A-2362-43 Slip-on† end (less gasket) x flange end
- ☐ Sizes – 4", 6", 8", 10", 12"
- ☐ Meets or exceeds all applicable requirements of ANSI/AWWA C509 Standard, UL Listed, FM Approved, and certified to ANSI/NSF 61.
- ☐ Slip-on ends fit plain end of classes 150, 200 and 250 cast iron, ductile iron and classes 150 and 200 cast iron O.D. PVC
- ☐ Flange end dimensions and drilling comply with ANSI B16.1, class 125
- ☐ Iron body with nominal 10 mils MUELLER® Pro-Gard™ Epoxy Fusion Coated interior and exterior surfaces
- ☐ Epoxy coating meets or exceeds all applicable requirements of ANSI/AWWA C550 Standard and is certified to ANSI/NSF 61
- ☐ Iron wedge, symmetrical & fully encapsulated with molded rubber; no exposed iron
- ☐ Non-rising stem (NRS)
- ☐ Triple O-ring seal stuffing box (2 upper & 1 lower O-rings), with fourth O-ring serving as dirt seal
- ☐ 2" square wrench nut (optional handwheel available) – open left or open right
- ☐ 250 psig (1723 kPa) maximum working pressure, 500 psig (3447 kPa) static test pressure
- ☐ UL Listed, FM Approved: 200 psig (1379 kPa)



A-2362-41

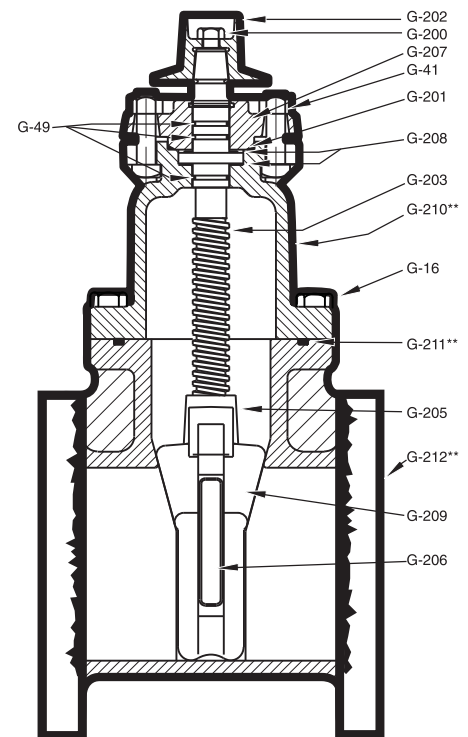
†Design and dimensions of the joint are manufactured under license of U.S. Pipe and Foundry Company.

Options

- ☐ Position indicators
- ☐ Handwheel

Resilient wedge gate valve parts

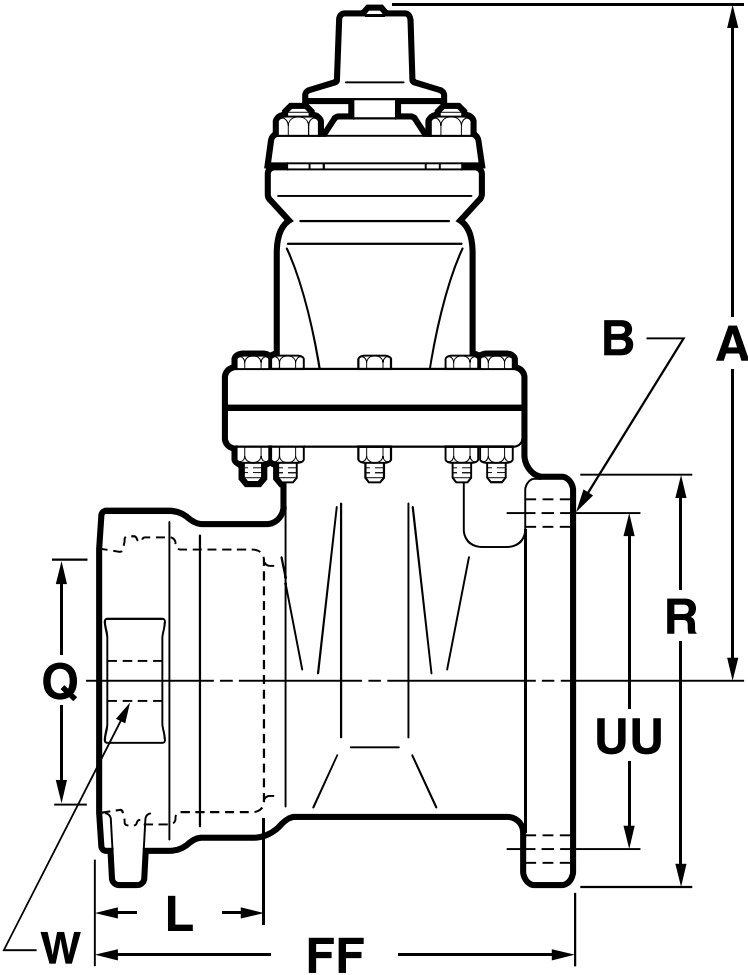
Catalog Part No.	Description	Material	Material standard
G-16	Bonnet Bolts & Nuts	Stainless Steel	Type 316
G-41	Stuffing Box Bolts & Nuts	Stainless Steel	Type 316
G-49	Stem O-rings (3)	EPDM	
G-200	Wrench Nut Cap Screw	Stainless Steel	Type 316
G-201	Stuffing Box Seal	EPDM	
G-202	Wrench Nut	Cast Iron	ASTM A126 CL.B
G-203	Stem	Everdur	ASTM B98
G-204	Hand Wheel (not shown)	Cast Iron	ASTM A126 CL.B
G-205	Stem Nut	Bronze	ASTM B62
G-206	Guide Cap Bearings	Celcon	
G-207	Stuffing Box with dirt seal	Cast iron Rubber	ASTM A126 CL.B --
G-208	Anti-friction Washers (2)	Celcon	
G-209	Wedge, Rubber Encapsulation	Cast Iron* EPDM	ASTM A126 CL.B --
G-210**	Bonnet	Ductile Iron	ASTM A536 Grade 65-45-12
G-211**	Bonnet O-ring	EPDM	
G-212**	Body	Ductile Iron	ASTM A536 Grade 65-45-12



* Fully encapsulated in molded rubber with no iron exposed

4"-12" A-2362-E381 RESILIENT WEDGE GATE VALVE - SL. x FL.

Mueller Co.



Dimensions

Dimension*	Nominal size				
	4"	6"	8"	10"	12"
A	14.19	18.00	21.50	25.50	28.62
Q (bore)	4.30	6.30	8.30	10.30	12.30
L	4.12	4.38	5.62	5.62	5.62
R	9.00	11.00	13.50	16.00	19.00
UU (bolt circle diameter for FL)	7.50	9.50	11.75	14.25	17.00
B (number and size of holes for FL)	8--.75	8--.88	8--.88	12--1.00	12--1.00
W (distance between strapping lug holes across face of valve ends)	8.62	10.62	13.12	16.38	18.12
FF	10.91	12.66	14.69	15.72	16.31
Turns to open	14	20.5	26.5	33	38.5
Weight*	90	165	260	400	565

*All dimensions are in inches. All weights are in pounds and are approximate.